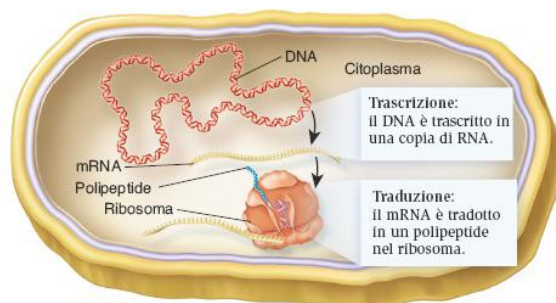


The code of Life: La Traduzione degli mRNA in Proteine

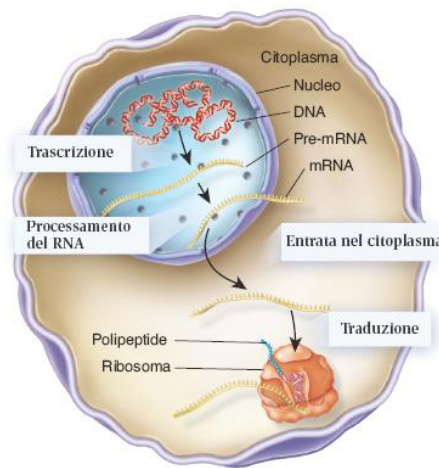
In eukaryotes:

-**transcription** occurs in the **nucleus**

-**translation** occurs in the **cytoplasm**

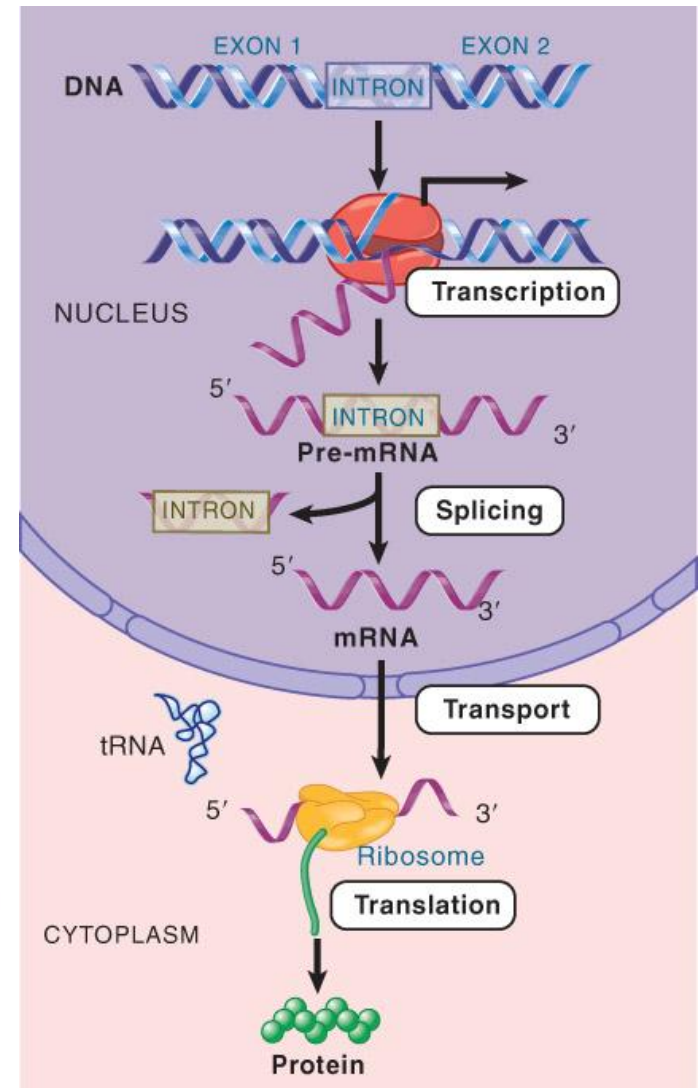


(a) Espressione genica molecolare nei procarioti



(b) Espressione genica molecolare negli eucarioti

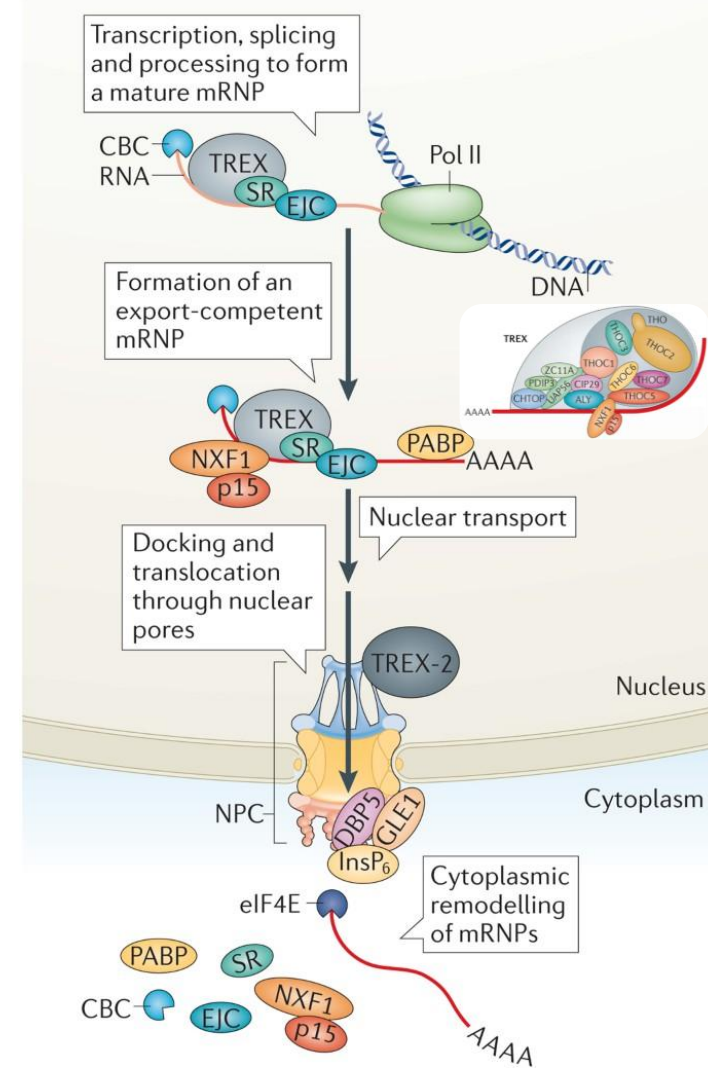
Figura 12.3 Dogma centrale dell'espressione genica a livello molecolare. (a) Nei batteri che sono privi del nucleo cellulare, la trascrizione e la traduzione avvengono nel citoplasma. (b) Negli eucarioti, la trascrizione e il processamento del DNA avvengono nel nucleo, mentre la traduzione avviene nel citoplasma.



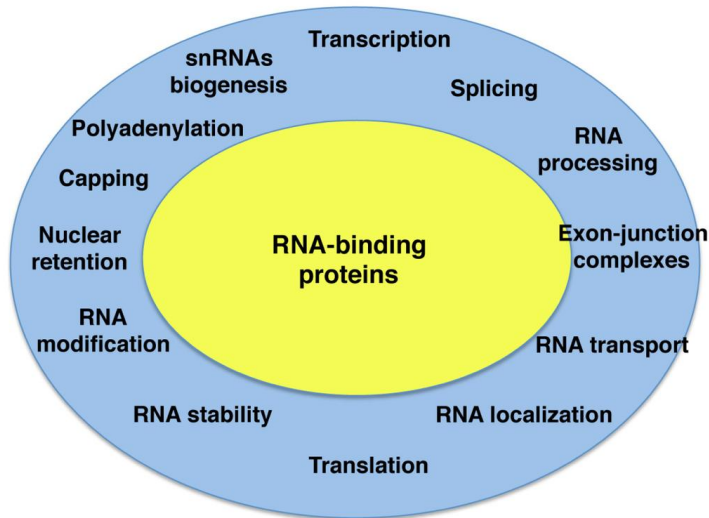
The Code of Life: Dal Nucleo al Citoplasma

Messenger Ribonucleoprotein Particles (mRNPs)

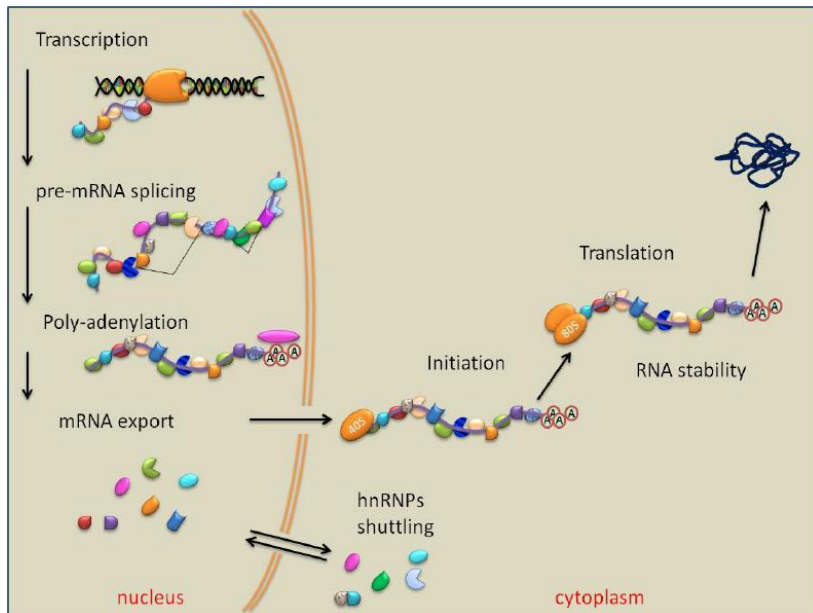
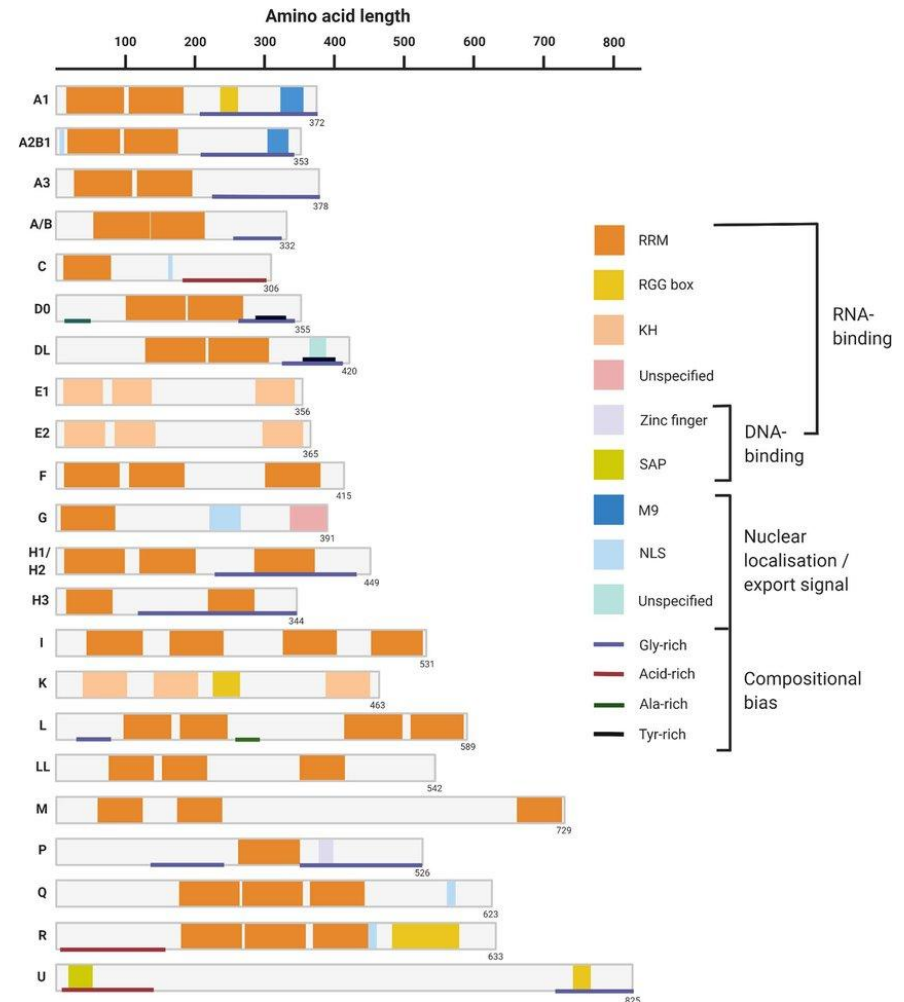
- mRNA export is an important node in regulation of mammalian gene expression.
- The cap structure (m7G) recruits the cap-binding complex (CBC) and induces RNA nuclear export.
- Two conserved complexes have key roles in the nuclear export of mRNA: transcription-export complexes TREX and TREX-2.
- Several recent examples indicate that specific biological pathways can be regulated by selective mRNA export, giving priority to some sets of mRNA over others.
- Nuclear export of functionally related sets of transcripts may be regulated by specific mRNA-binding export factors that recognize specific sequence elements among the mRNAs.



The Code of Life: Proteine Leganti gli RNA

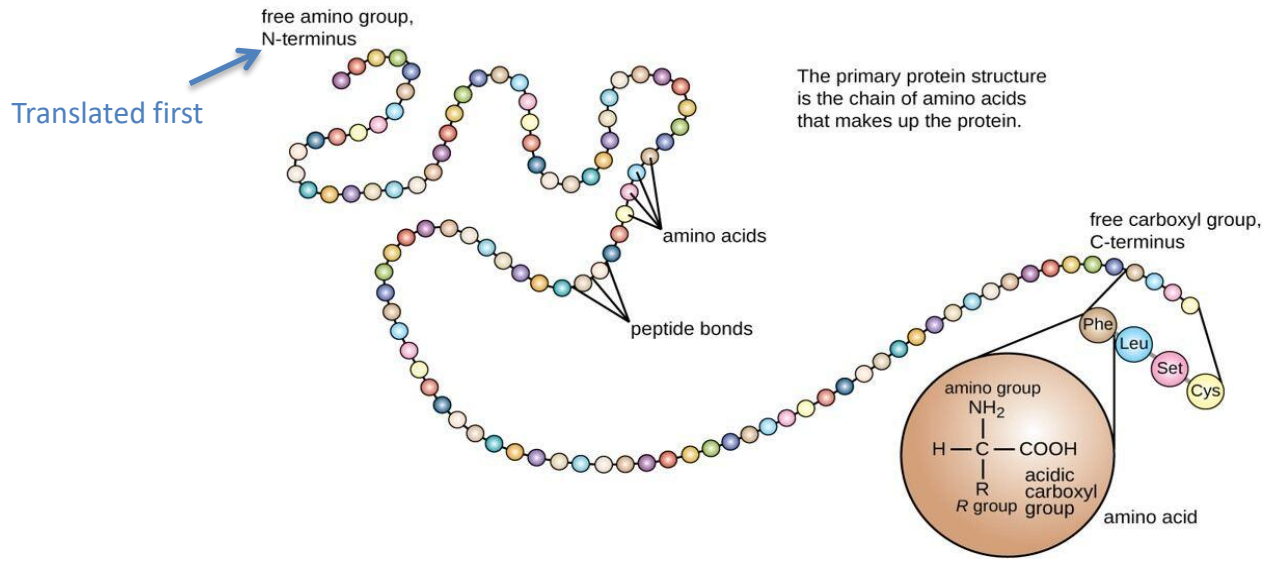


Heterogeneous nuclear ribonucleoproteins (hnRNPs)

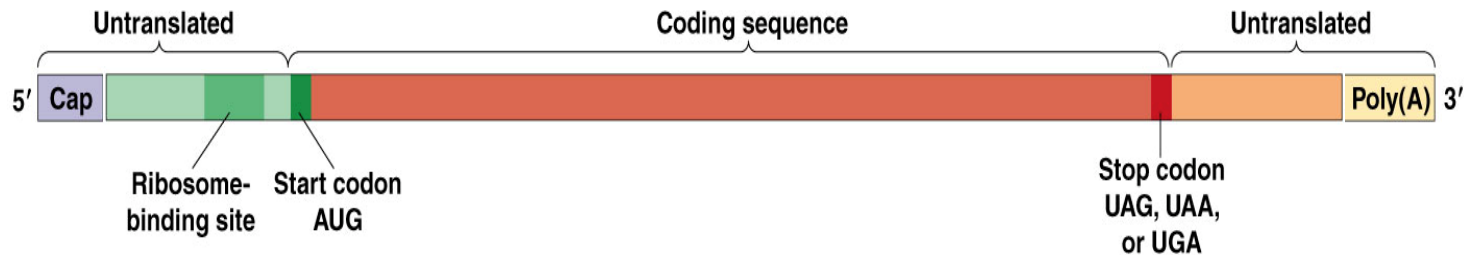


The Mechanism of Translation

- Translation is an ordered, stepwise process that **begins at the N-terminus of the polypeptide and adds amino acids to the growing chain until the C-terminus is reached**



- The **mRNA** is read in the **5' to 3' direction**

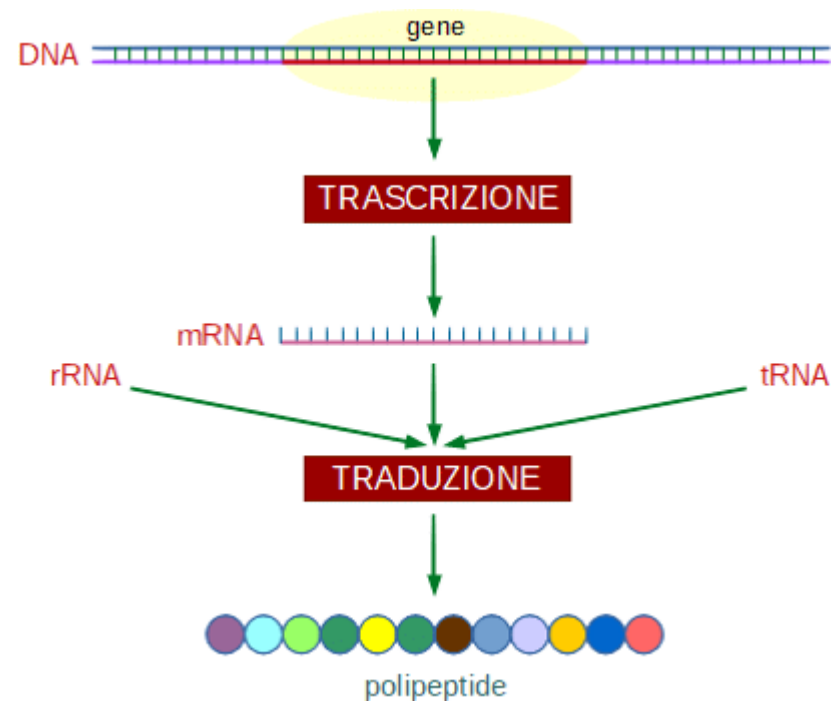


The Code of Life: La Traduzione

Da una sequenza di nucleotidi ad una sequenza di amino acidi

A large number of components are required for the synthesis of a protein:

- **mRNA** to be translated
- **Amino acids**
- Transfer RNA (**tRNA**)
- **Ribosomes**
- **Energy** sources
- **Proteins** needed for **initiation, elongation, and termination** of the polypeptide chain.



Sequenza del Gene BRCA1

UPPER CASE RED = Sequenze regolatorie al 5' e 3' del mRNA di lunghezza variabile facenti parte dei primi ed ultimi ESONI ma che non vengono tradotte in sequenze aminoacidiche = UTR.

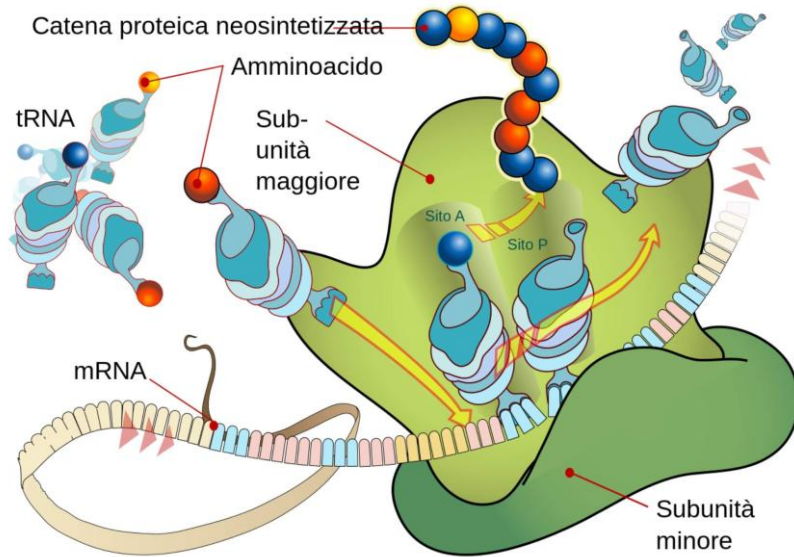
UPPER CASE BLUE = sequenza nucleotidica facente parte degli ESONI che viene tradotta nella sequenza aminoacidica.

lower case grey = sequenza nucleotidica facente parte degli INTRONI che NON viene tradotta nella sequenza aminoacidica.

No.	Exon / Intron	Length	Sequence
	5' upstream sequence		tgatttccattcttgagaggctattgtctcagcggtttccgtggcaacagt
1	ENSE00003791262	120	AAAGCGTGGGAATTACAGATAAAATAAAACGTGGAACCCCTTTCCTCGGTGCGGCCAA GGTGTTCGTCTCCAGGGAAGCTAAGGCCGCTTGGGGTGAGACCTCACTTCATCCG
	Intron 1-2	46,010	gtgagtagcaccgcgtccggcagcc.....tatatgtttttctaagtgtttaaag
2	ENSE00003559512	99	TTCATTGGAACAGAAAGAAATGGATTATCTGCTCTTCGCGTTGAAGAAGTACAAAATGT CATTAAATGCTATGCAGAAAATCTTAGAGTGTCCCATCTG
	Intron 2-3	8,237	gtaagtgcagcacaagagtgtattaa.....tttttctccccctaccctgtctag
3	ENSE00003510592	54	TCTGGAGTTGATCAAGGAACCTGTCTCCACAAGTGTGACCACATATTTTGC
	Intron 3-4	9,192	gtaagtttgaatgtgttatgtggt.....tcttcttcttcttataatttatag
4	ENSE00003541068	78	ATTTTGCATGCTGAACCTCTCACCAGAAAGGGCCCTTCACAGTGTCTTTATGTAA GAATGATATAACCAAAAG
	Intron 4-5	1,499	gtatataatttggttaagtgtctag.....tgtttctcaacaatttaatttcag
5	ENSE00003531836	89	GAGCCTACAAGAAAGTACGAGATTAGTCAACTTGTGAAGAGCTATTGAAATCATTG TGCTTTTCAGCTTGACACAGTTTGAGT
	Intron 5-6	606	gtaagtggttaatatcccaagaatg.....taagtgttttccctgtattttacag
6	ENSE00001917948	140	ATGCAACAGCTATAATTTTGC AAAAGGAAATAACTCTCTGAACATCTAAAGATG AAGTTTCTATCATCAAGATATGGGCTACAGAAACCTGCCAAAAGACTTCTACAGAGT AACCAGAAATCCTTCCTTG
	Intron 19-20	1,417	gtaagtattgggtgcccgtgcagag.....tgctcttcttcttctgtgggatccag
20	ENSE00004011552	61	GGTGTCCACCAATTTGTGGTTGTGCAGCCAGATGCCTGGACAGAGACAATGGCTTCCAT G
	Intron 20-21	1,840	gtaaggtgctgcatgtacctgtgc.....ctctgctgtgttctctgtctccag
21	ENSE00004011564	1,508	CAATTGGGCAGATGTGTGAGGCACCTGTGGTGACCCGAGAGTGGGTGTGGACAGTGTAG CACTCTACCAAGTGCAGAGAGTGGACACCTACCTGATACCCAGATCCCCACAGCCACT ACTGACTGCAGCCAGCCACAGGTACAGAGCCACAGGACCCCAAGAAATGAGCTTACAAAGT GGCCTTCCAGGCCCTGGGAGCTCCTCTCACTCTTCAGTCTCTACTGTCTCTGGCTACT AAATATTTTATGTACATCAGCCTGAAAAGGACTTCTGGCTATGCAAGGGTCCCTTAAAGA TTTTCTGCTTGAAGTCTCCCTTGGAATCTGCCATGAGCACAAAATATGGTAATTTTC ACCTGAGAAGATTTTAAACCAATTTAAACGCCCAATTTGAGCAAGATGCTGATTCTTA TTTATCAGCCCTATTCTTTCTATTTCAGGCTGTTGTTGGCTTAGGGCTGGAAGCACAGAGT GGCTTGGCCTCAAGAGAATAGCTGGTTCCCTAAGTTTACTTCTCTAAACCCCTGTGTTTC ACAAAGGCAGAGATCAGACCCCTTCAATGGAAGGAGAGTGTGGGATCGATTATGTGAC TTAAAGTCAGAAATAGTCTTGGGAGTCTCAAAATGTTGGAGTGAACATTGGGGAGGA ATTCTGAGGCAGGTATTAGAAATGAAAAGGAACTTGAACCTGGGCATGTTGGCTCAGC CCTGTAATCCAGCACTTTGGGAGGCCAAGTGGGAGATCACTGGAGGTGAGGATTTGC AAACAGCCTGGCCACATGGTGAAACCCCTCTCTACTAAATACAGAAATAGCCCGG TCATGGTGGTGACACCTGTAATCCAGCTACTCAGGTGGCTAAGGCAGGAGAATCACTT CAGCCCGGAGGTGGAGTTGCAGTGAGCCAGATCATACCGGCACTCCAGCCTGGGT GACAGTGAGACTGTGGCTCAAAAAAAAAAAAAAAAAAGGAAATGAACTAGAAAGAGAT TTCTAAAGTCTGAGATATATTTGTAGATTCTTAAAGAATGTGTTCTAAACAGCAGAA GATTTTCAAGAACCGTTTCCAAAGACAGTCTTCTAATTCCTCATTAGTAATAAGTAAAA TGTTTATTTGTTAGTCTCTGGTATATAATCCATTCTCTTAAATATAAGCACTCTGGCA TGAATATTTCTATCTATAAATGACAGATCCACAGGAAGGAAGCTGTGCTTTCTTTT GAGGTGATTTTTTCTTTGCTCCCTGTGTCTGAAACCATACAGCTTCATAAATAATTTT GCTTGTCTGAAGGAAGAAAAGTGTTTTCTATAAACCCATATCCAGGACTGTTTATAGCT GTTGGAAGGACTAGGTCTTCCCTAGCCCCCAGTGTGCAAGGGCAGTGAAGACTGTATT GTACAAATACGTTTTGTAATGTTGTCTGTTAACTGCAATAAACTTGGTAGCAAA CACTTCCA
	3' downstream sequence		ccatgaatgactgttcttgagacttaggcagccgactttctcagagcct.....



Dalla Sequenza dell'mRNA Alla Sequenza amino acidica di BRCA1



TRE nucleotidi = 1 Aminoacido
 ATG (AUG) = Codone di INIZIO = Metionina
 TGA (UGA) = Codone di STOP

1	AAAGCGTGGGAATTACAGATAAAATAAAACTGTGGAACCCCTTTCTCGGCTGCCGCCAA	60
61	GGTGTTCCGGTCCTTCCGAGGAAGCTAAGGCCGCGTTGGGGTGAGACCCTCACTTCATCCG	120
121	TTCATTGGAACAGAAAGAAATGGATTTATCTGCTCTTCGCGTTGAAGAAGTACAAAATGT	180
42	CATTAATGCTATGCAGAAAATCTTAGAGTGTCCCATCTGTCTGGAGTTGATCAAGGAACC	101
14	--I--N--A--M--Q--K--I--L--E--C--P--I--C--L--E--L--I--K--E--P	34
241	TGCTCCACAAAGTGTGACCACATATTTTGCAAATTTGTCATGCTGAAACTTCTCAACCA	300
102	TGCTCCACAAAGTGTGACCACATATTTTGCAAATTTGTCATGCTGAAACTTCTCAACCA	161
34	--V--S--T--K--C--D--H--I--F--C--K--F--C--M--L--K--L--L--N--Q	54
301	GAAGAAAGGGCCTTCACAGTGTCTTTATGTAAGAATGATATAACCAAAGGAGCCTACA	360
162	GAAGAAAGGGCCTTCACAGTGTCTTTATGTAAGAATGATATAACCAAAGGAGCCTACA	221
54	--K--K--G--P--S--Q--C--P--L--C--K--N--D--I--T--K--R--S--L--Q	74
361	AGAAAGTACGAGATTTAGTCAACTTGTTGAAGAGCTATTGAAAATCATTTGTGCTTTTCA	420
222	AGAAAGTACGAGATTTAGTCAACTTGTTGAAGAGCTATTGAAAATCATTTGTGCTTTTCA	281
74	--E--S--T--R--F--S--Q--L--V--E--E--L--L--K--I--I--C--A--F--Q	94
421	GCTTGACACAGGTTTGGAGTATGCAACAGCTATAATTTTGCAAAAAGGAAAATAACTC	480
282	GCTTGACACAGGTTTGGAGTATGCAACAGCTATAATTTTGCAAAAAGGAAAATAACTC	341
94	--L--D--T--G--L--E--Y--A--N--S--Y--N--F--A--K--K--E--N--N--S	114
↓ ↓ ↓ ↓		
5461	GACAGAGGACAATGGCTTCCATGCAATTGGGCAGATGTGTGAGGCACCTGTGGTGACCCG	5520
5322	GACAGAGGACAATGGCTTCCATGCAATTGGGCAGATGTGTGAGGCACCTGTGGTGACCCG	5381
1774	--T--E--D--N--G--F--H--A--I--G--Q--M--C--E--A--P--V--V--T--R	1794
5521	AGAGTGGGTGTTGGACAGTGTAGCACTCTACCAGTGCCAGGAGCTGGACACCTACCTGAT	5580
5382	AGAGTGGGTGTTGGACAGTGTAGCACTCTACCAGTGCCAGGAGCTGGACACCTACCTGAT	5441
1794	--E--W--V--L--D--S--V--A--L--Y--Q--C--Q--E--L--D--T--Y--L--I	1814
5581	ACCCAGATCCCCACAGCCACTACTGACTGTCAGCCAGCCACAGGTACAGAGCCACAGGA	5640
5442	ACCCAGATCCCCACAGCCACTACTGACTGACTGTCAGCCAGCCACAGGTACAGAGCCACAGGA	5469
1814	--P--Q--I--P--H--S--H--Y--*	1822
5641	CCCCAAGAATGAGCTTACAAAGTGGCCTTTCCAGGCCCTGGGAGCTCCTCTCACTCTTCA	5700

The Code of Life: il Codice Genetico

Da una sequenza di nucleotidi ad una sequenza di amino acidi

3 nucleotidi = 1 codone, 64 codoni, 20 amino acidi

	GCA	AGA AGG CGA CGC CGG CGU	GAC GAU	AAC AAU	UGC UGU	GAA GAG	CAA CAG	GGA GGC GGG GGU	CAC CAU	AUA AUC AUU	UUA UUG CUA CUC CUG CUU	Inizio ↓ AUG	AAA AAG	UUC UUU	CCA CCC CCG CCU	AGC AGU UCA UCC UCG UCU	ACA ACC ACG ACU	UGG	UAC UAU	GUA GUC GUG GUU	UAA UAG UGA
Nucleotidi	Ala	Arg	Asp	Asn	Cys	Glu	Gln	Gly	His	Ile	Leu	Lys	Met	Phe	Pro	Ser	Thr	Trp	Tyr	Val	stop
Amino acidi	A	R	D	N	C	E	Q	G	H	I	L	K	M	F	P	S	T	W	Y	V	

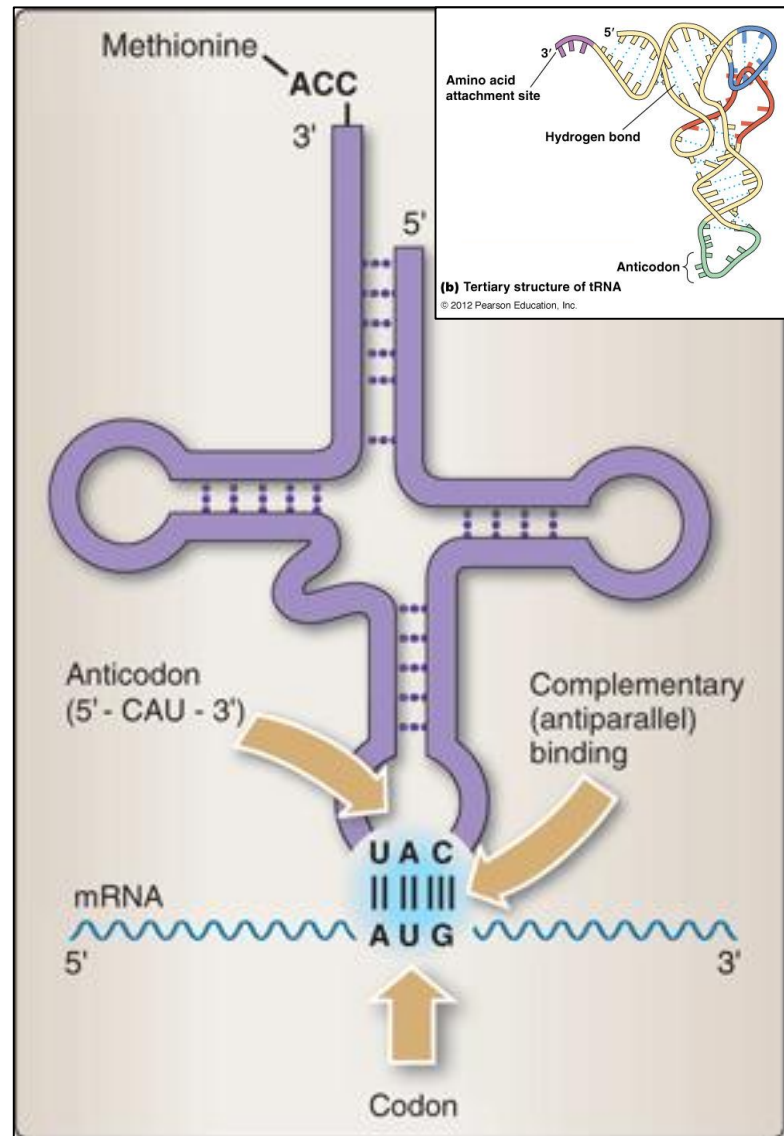
Figure 6-48 Molecular Biology of the Cell 6e (© Garland Science 2015)

Characteristics of the genetic code include:

- The genetic code is **read from a fixed starting point as a continuous sequence** of bases, taken three at a time (codon).
- The genetic code is **specific (unambiguous)**, that is, a particular codon always codes for the same amino acid.
- The genetic code is degenerate (**sometimes called redundant**). Although each codon corresponds to a single amino acid, a given amino acid may have more than one codon coding for it. For example, arginine is specified by six different codons. Multiple codons that represent the same amino acid **most often differ at the third base position** (see Alanine).

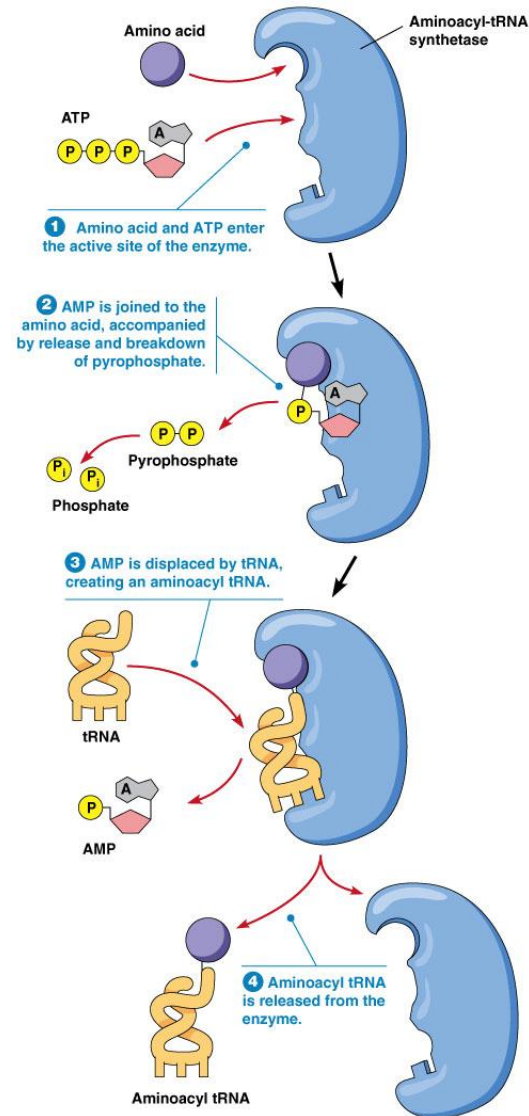
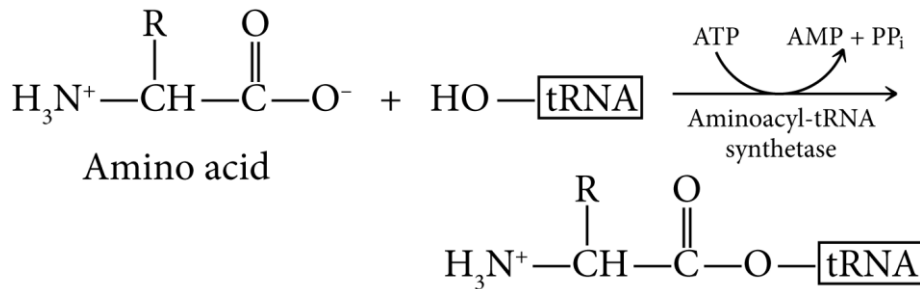
The Code of Life: tRNA la molecola convertitrice

- **tRNAs (RNAs transfer)** are able to **carry a specific amino acid** and to **recognize the relative codon for that amino acid**. tRNA, therefore, functions as adaptor molecules.
- **At least one specific type of tRNA is required per amino acid**. In humans, there are **almost 50** species of tRNA. Because there are only **20 different amino acids** commonly carried by tRNA, amino acids that are encoded by several codons have more than one specific tRNA molecule.



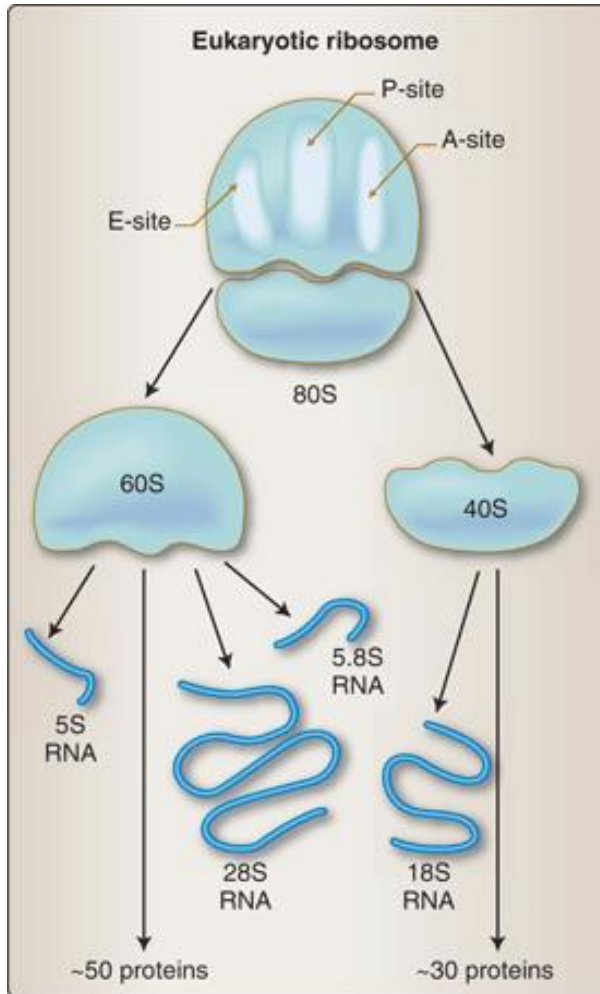
The Code of Life: Aminoacyl-tRNA synthesis

- **Aminoacyl-tRNA Synthetases** link amino acids to the correct Transfer RNAs
- There is **one aminoacyl-tRNA synthetase** for **each amino acid** → cells typically have **20 different aminoacyl-tRNA synthetases** to attach each amino acid to the appropriate tRNA
- Aminoacyl-tRNA synthetases catalyze the attachment of amino acids to the tRNAs via a multistep reaction involving **ATP hydrolysis**



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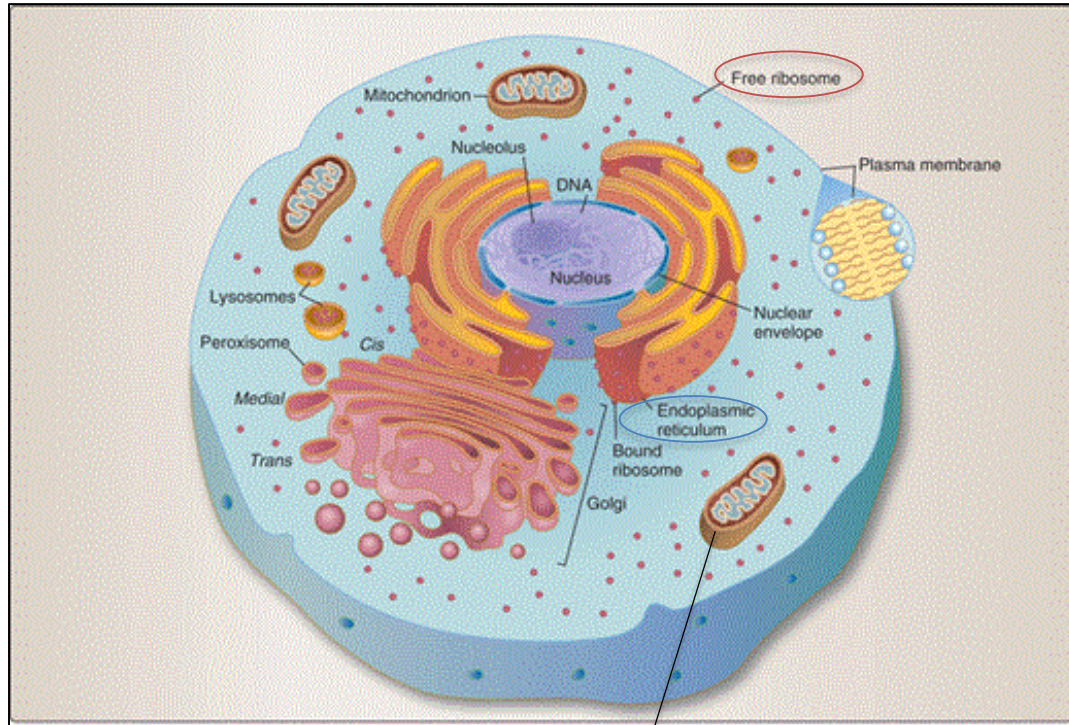
The Code of Life: Ribosomes



- **Ribosomes** are large **complexes of protein and ribosomal RNA (rRNA)**. They consist of **two subunits**—one large and one small—whose relative sizes are generally given in terms of their **sedimentation coefficients, or S (Svedberg)** values.
- The **larger ribosomal subunit catalyzes the formation of the peptide bonds** that link amino-acid residues in a sequence.
- The **smaller subunit binds mRNA** and is responsible for the accuracy of translation by **ensuring correct base-pairing** between the codon in the mRNA and the anticodon of the tRNA.

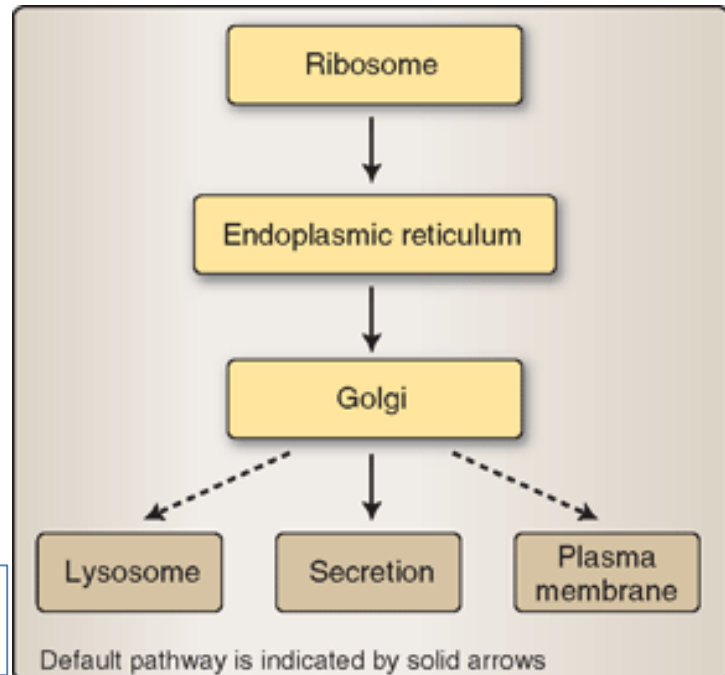
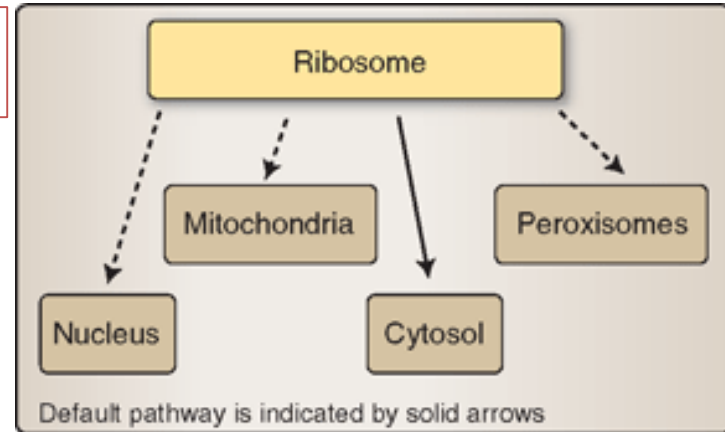
The Code of Life: Ribosomes localization

Free (cytoplasm) Ribosomes



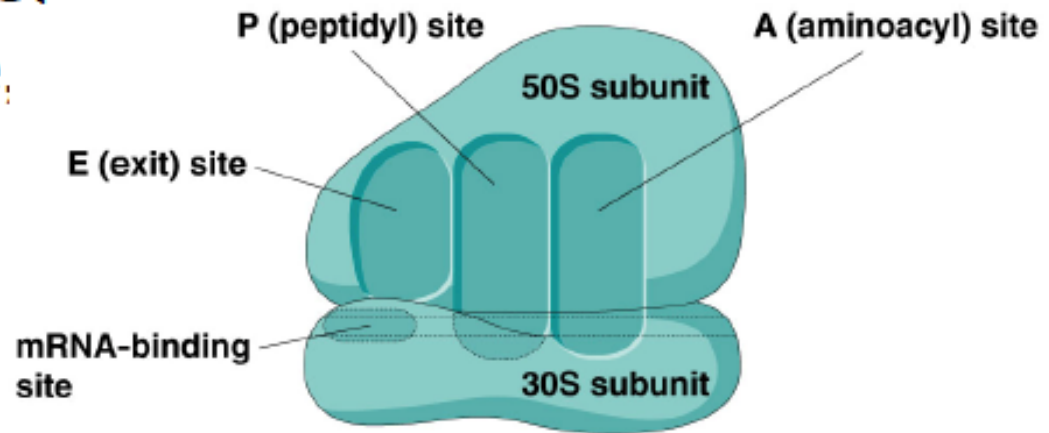
+ a subset of ribosome specifically present in mitochondria

ER-bound Ribosomes



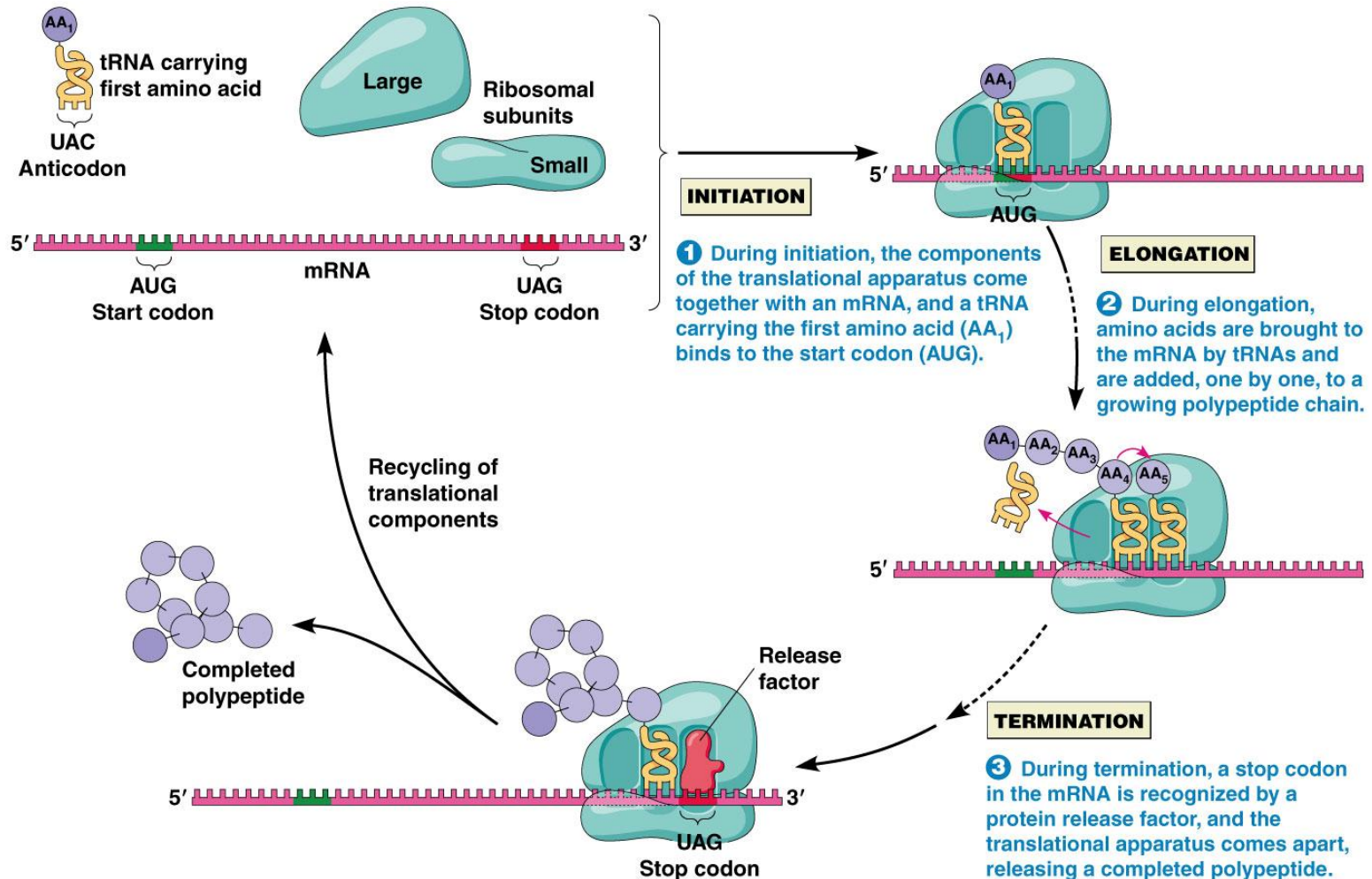
The Code of Life: Ribosomes

Ribosomes have four important sites:
the **mRNA binding site**,
the **A (aminoacyl) site**,
the **P (peptidyl) site**
the **E (exit) site**



The Code of Life: Phases of Translation

Translation is divided into three phases: initiation (1), elongation (2), and termination (3)



The Code of Life: Initiation

Initiation starts with the binding of 3 proteins (Initiation Factors 1, 2, 3) and 1 GTP to the small ribosomal subunit

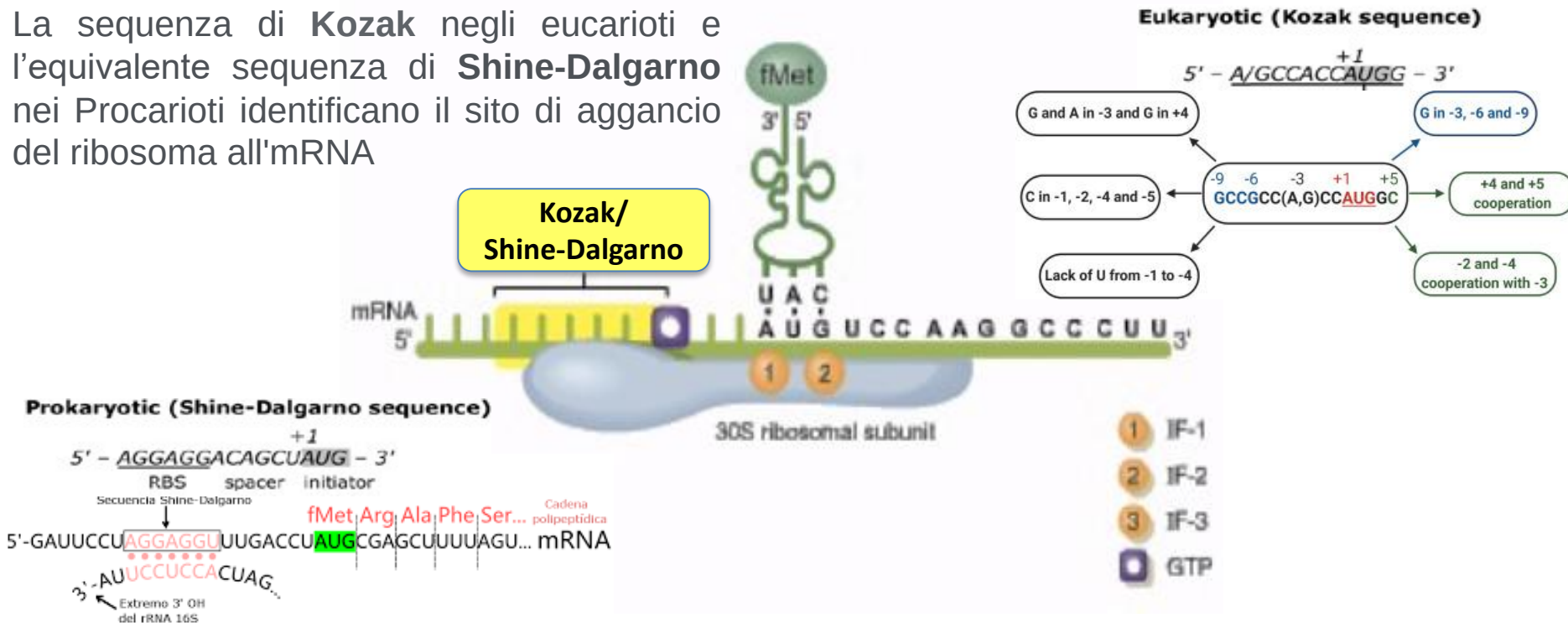
mRNA pair with the correct site of the small ribosomal subunit



The Code of Life: Initiation

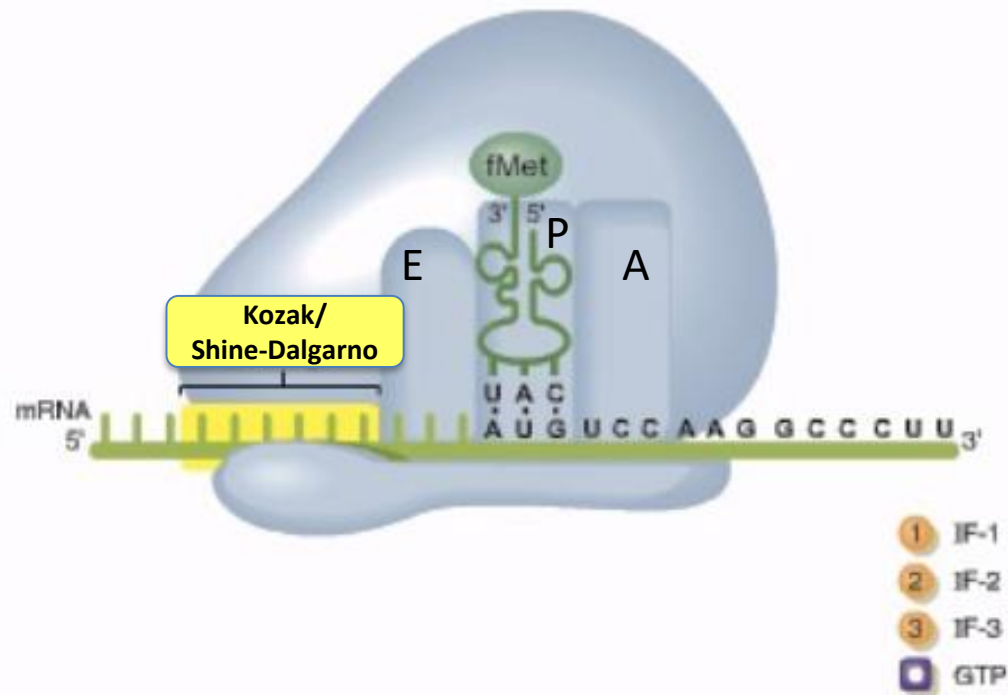
Translation begins always at the start codon AUG with a modified methionine in bacteria (formyl-Met), with a unmodified Met in Eukaryotes

La sequenza di **Kozak** negli eucarioti e l'equivalente sequenza di **Shine-Dalgarno** nei Procarioti identificano il sito di aggancio del ribosoma all'mRNA



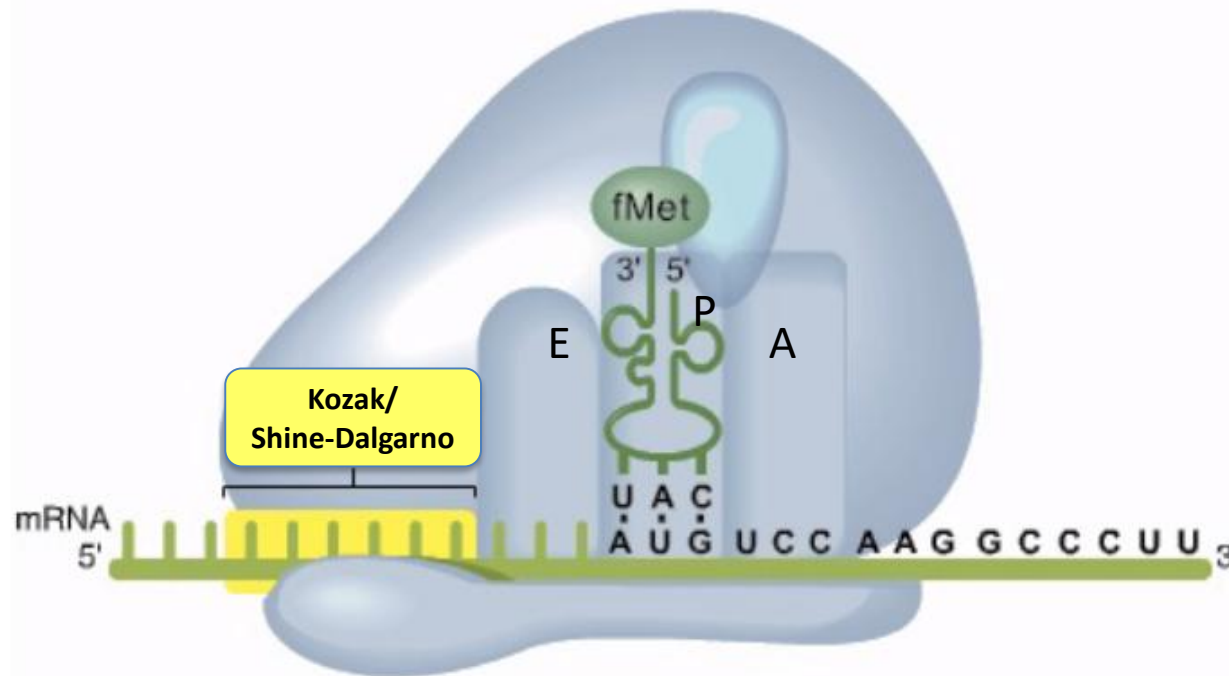
The Code of Life: Initiation

The last step in Initiation is the binding of the large ribosomal subunit, with the release of IF proteins and the (f)Met-tRNA sitting in the P site. Elongation starts.



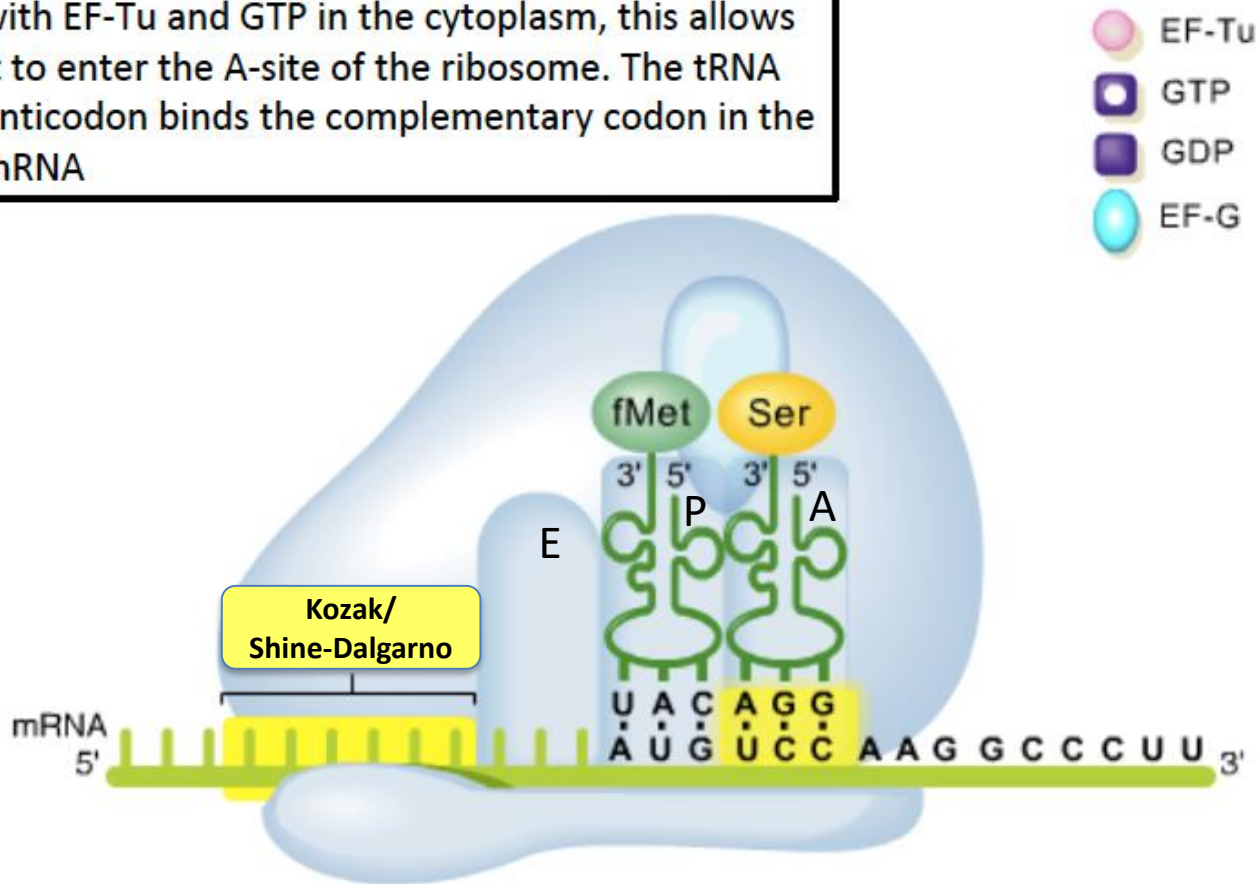
The Code of Life: Elongation

Elongation consists of 3 steps: binding of the next tRNA-AA, formation of the peptide bond, and sliding of the ribosome to the next position



The Code of Life: Elongation

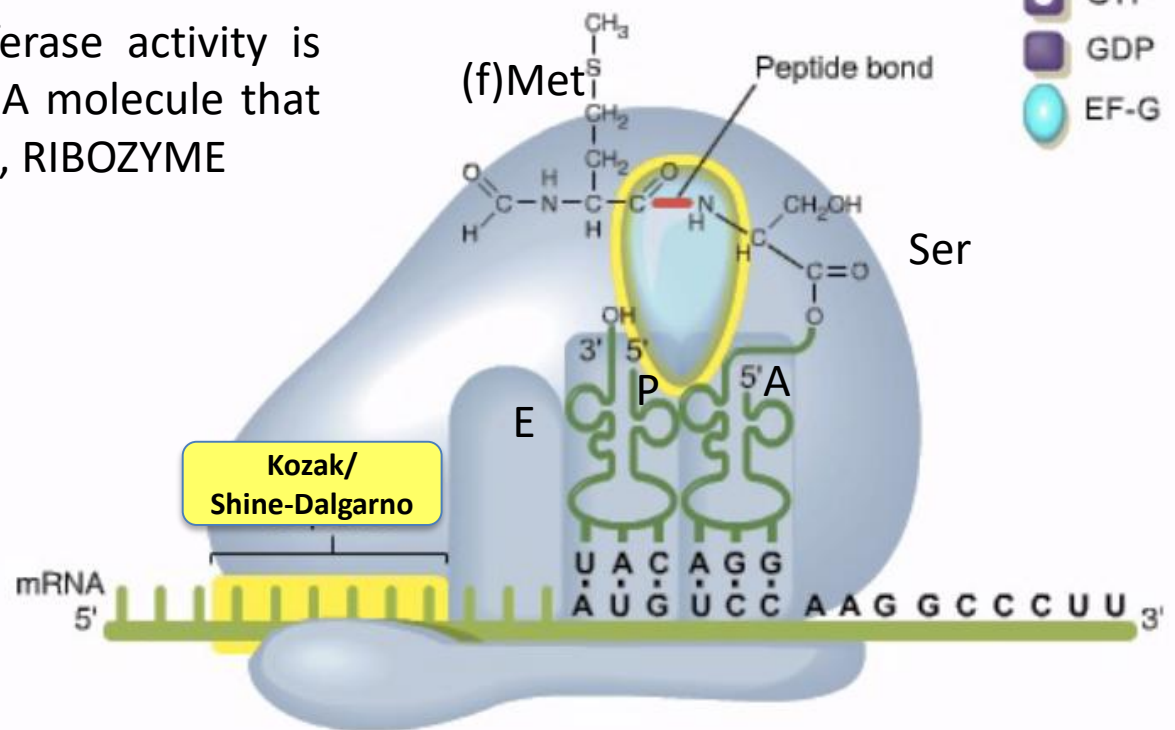
At the end of Initiation the Met-tRNA is in the P-site of the ribosome. The next tRNA complexes with EF-Tu and GTP in the cytoplasm, this allows it to enter the A-site of the ribosome. The tRNA anticodon binds the complementary codon in the mRNA



The Code of Life: Elongation

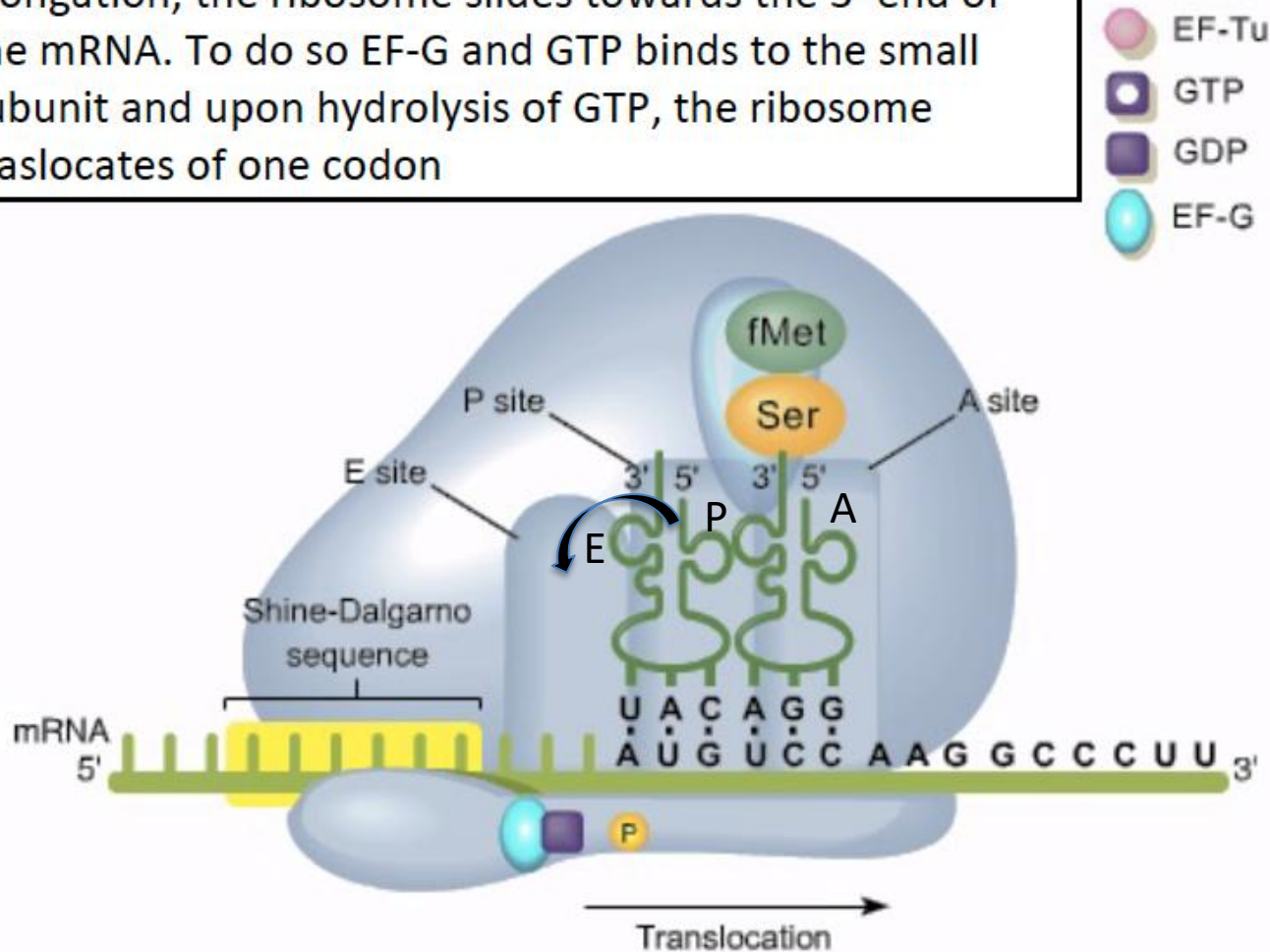
A peptide bond forms between the Met in the P-site and the Ser in the A site. This reaction is catalyzed by the peptidyl transferase activity of the large ribosomal subunit. This enzyme first detach the (f)Met from its tRNA and then condense the covalent binding between the 2 AA

The peptidyl transferase activity is carried out by a RNA molecule that works as an enzyme, RIBOZYME



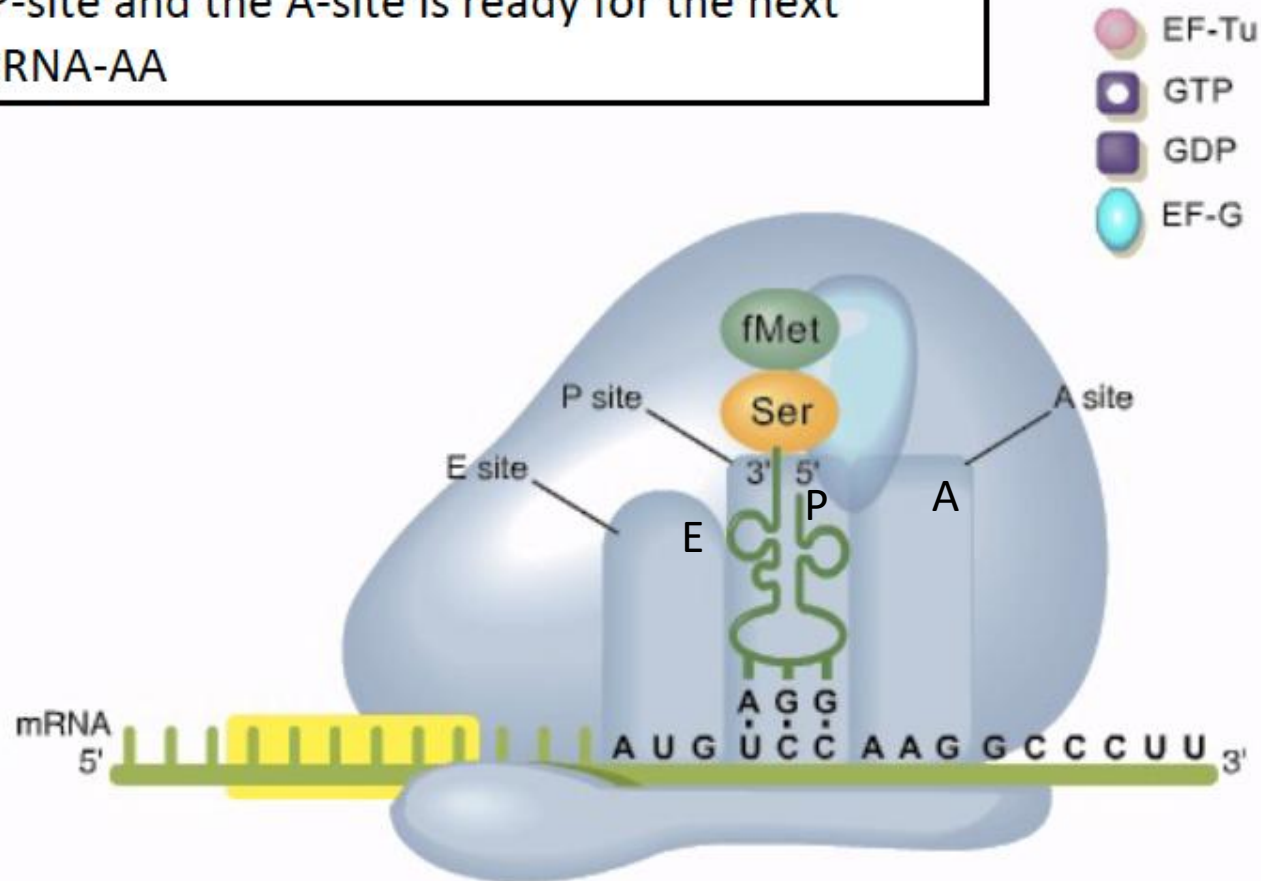
The Code of Life: Elongation

To remove the uncharged tRNA and progress with elongation, the ribosome slides towards the 3' end of the mRNA. To do so EF-G and GTP binds to the small subunit and upon hydrolysis of GTP, the ribosome translocates of one codon



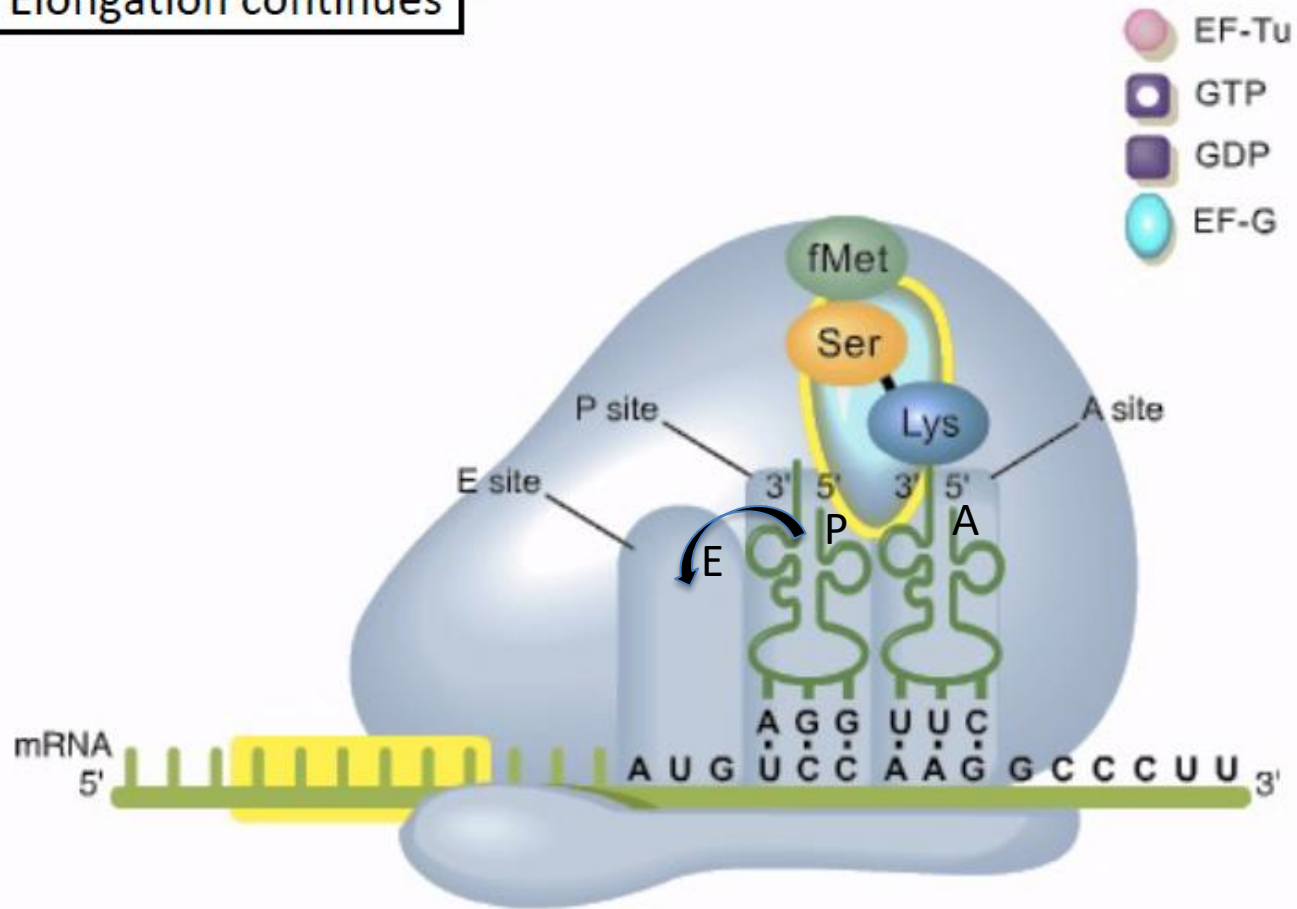
The Code of Life: Elongation

The sliding of the ribosome places the uncharged tRNA to the E site, the tRNA carrying the growing polypeptide moves to the P-site and the A-site is ready for the next tRNA-AA



The Code of Life: Elongation

Elongation continues



The Code of Life: Termination

Elongation continues till the ribosome reaches one of the 3 STOP codons

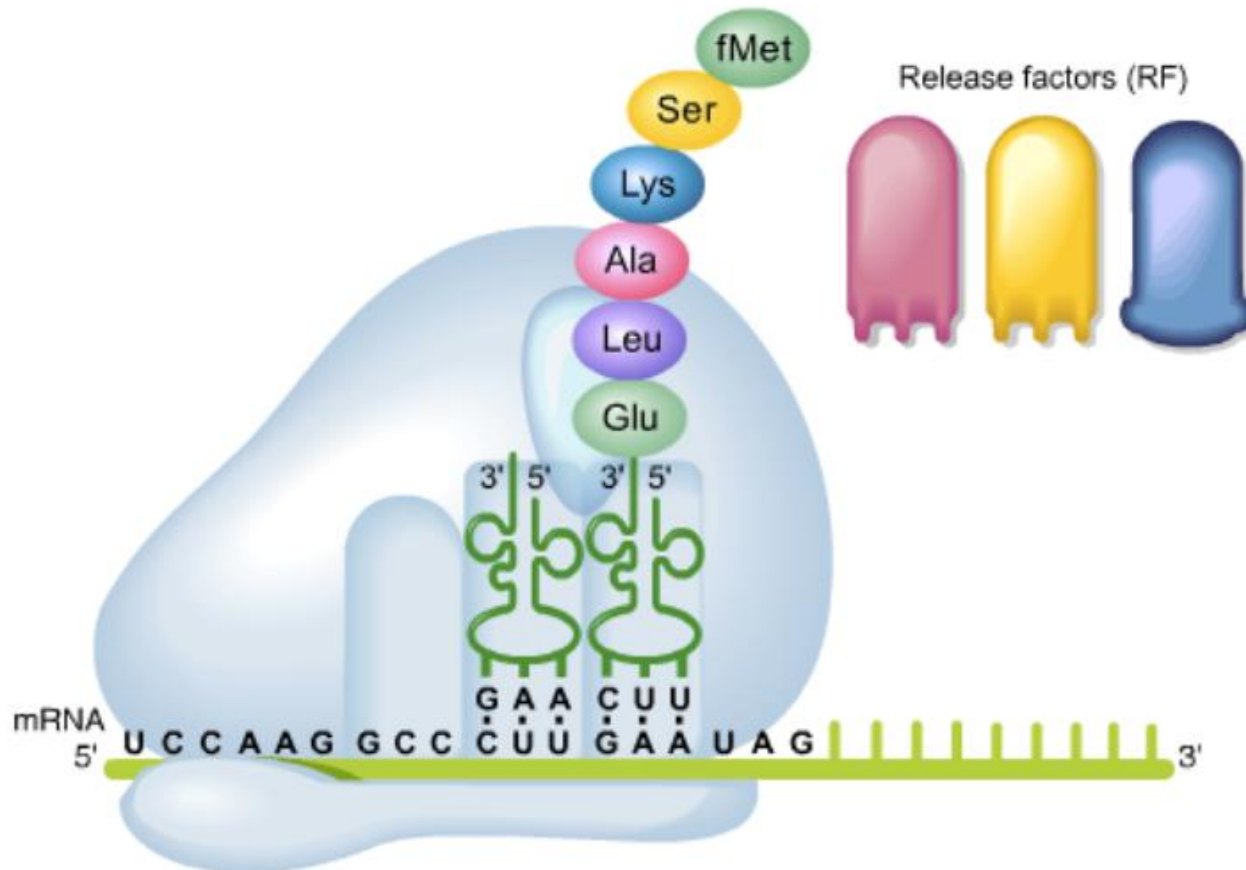
Termination

THE GENETIC CODE

		Second letter					
		U	C	A	G		
First letter	U	UUU Phe UUC (F) UUA Leu UUG (L)	UCU UCC Ser (S) UCA (S) UCG	UAU Tyr UAC (Y) UAA Stop UAG Stop	UGU Cys UGC (C) UGA Stop UGG Trp (W)	U C A G	Third letter
	C	CUU CUC Leu CUA (L) CUG	CCU CCC Pro CCA (P) CCG	CAU His CAC (H) CAA Gln CAG (Q)	CGU CGC Arg CGA (R) CGG	U C A G	
	A	AUU AUC Ile (I) AUA AUG Met (M)	ACU ACC Thr ACA (T) ACG	AAU Asn AAC (N) AAA Lys AAG (K)	AGU Ser AGC (S) AGA Arg AGG (R)	U C A G	
	G	GUU GUC Val GUA (V) GUG	GCU GCC Ala GCA (A) GCG	GAU Asp GAC (D) GAA Glu GAG (E)	GGU GGC Gly GGA (G) GGG	U C A G	

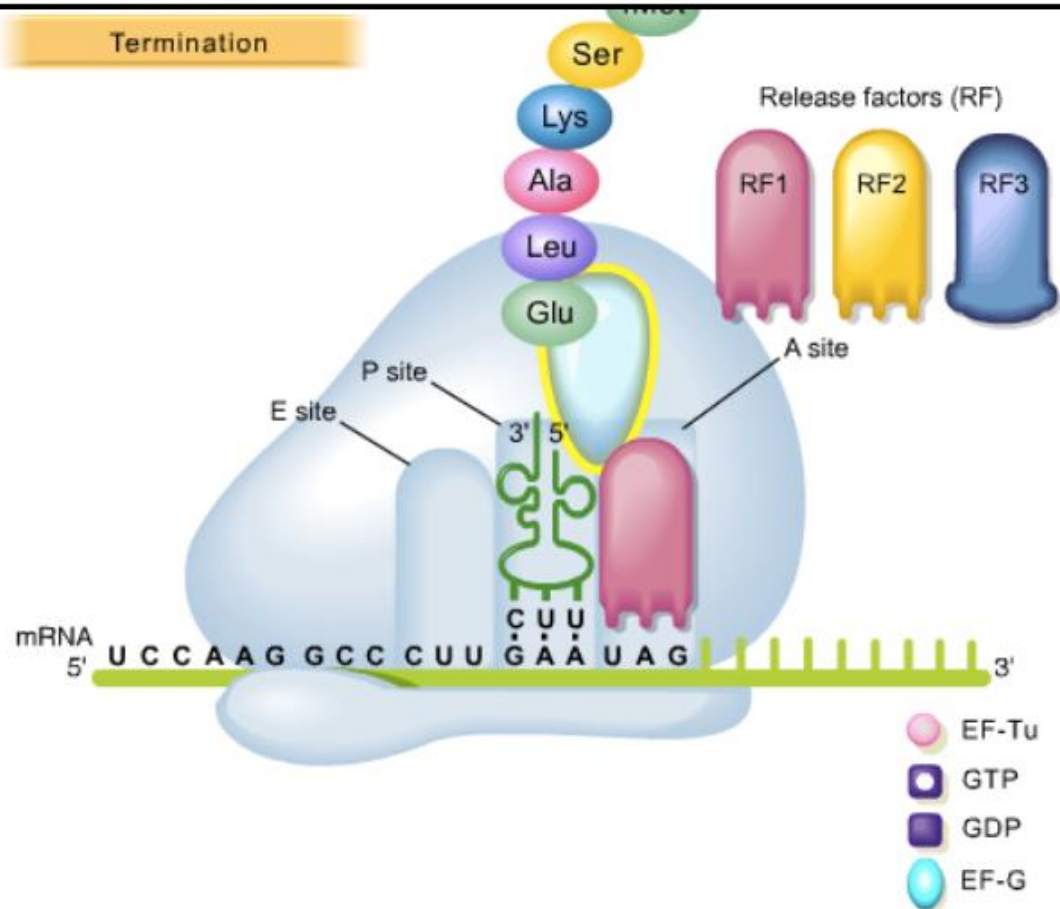
The Code of Life: Termination

There are no tRNAs for the stop codons. Instead, proteins, called Release Factors read the stop codons and initiate the termination events



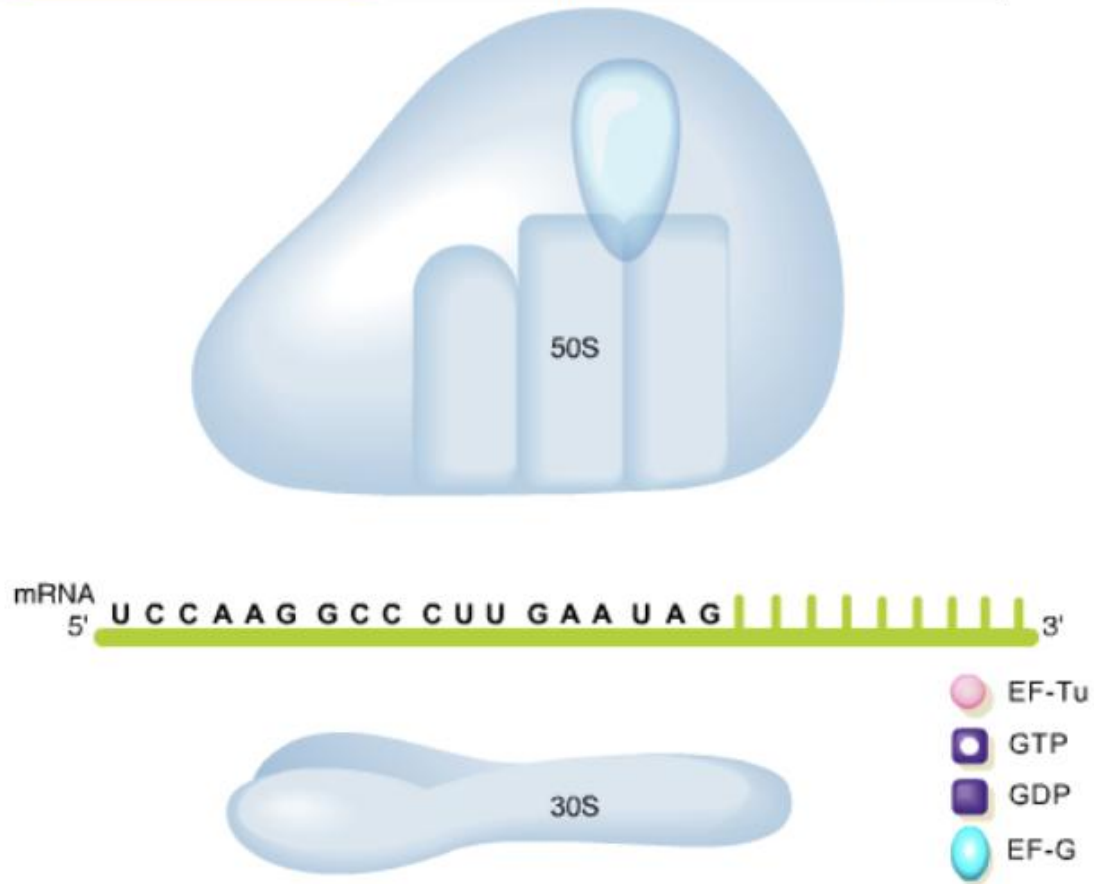
The Code of Life: Termination

When the stop codon is at the A-site, this is occupied by the corresponding RF. This triggers the release of the polypeptide from the last tRNA in the P-site

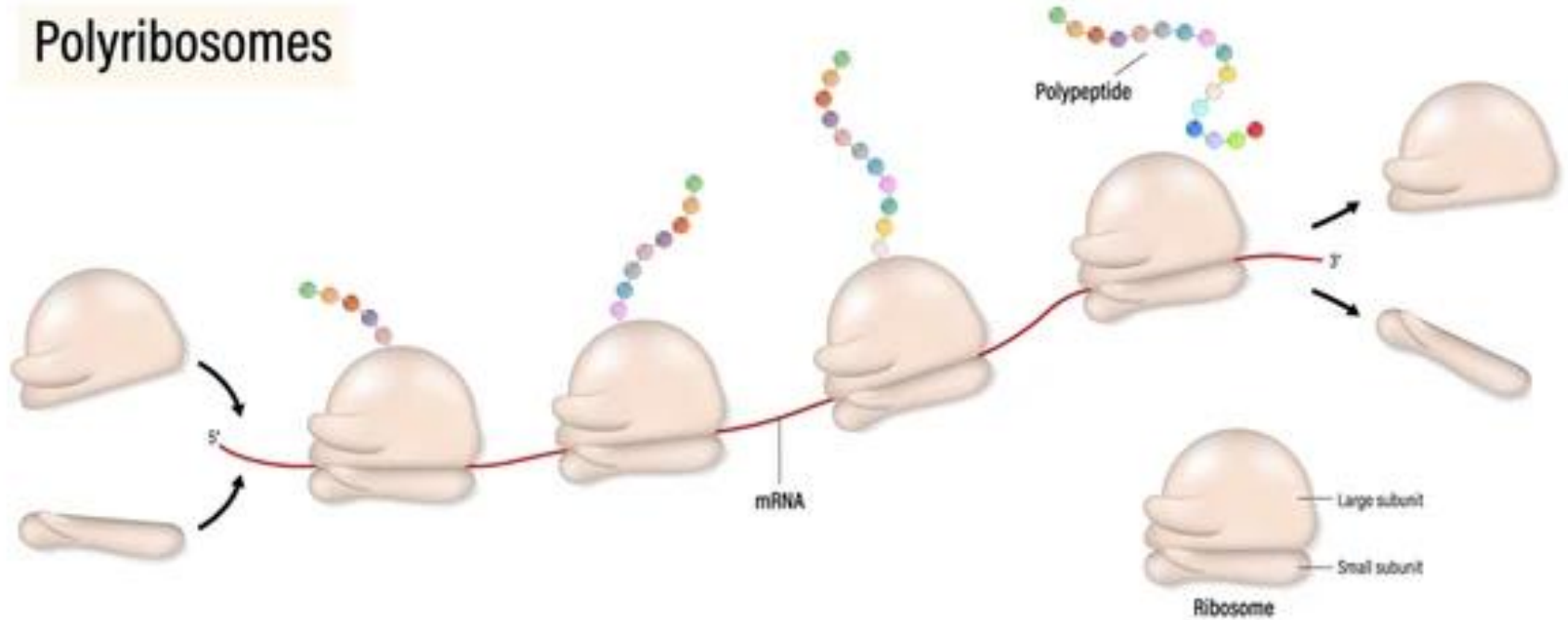


The Code of Life: Termination

Everything disassemble and the components are recycled

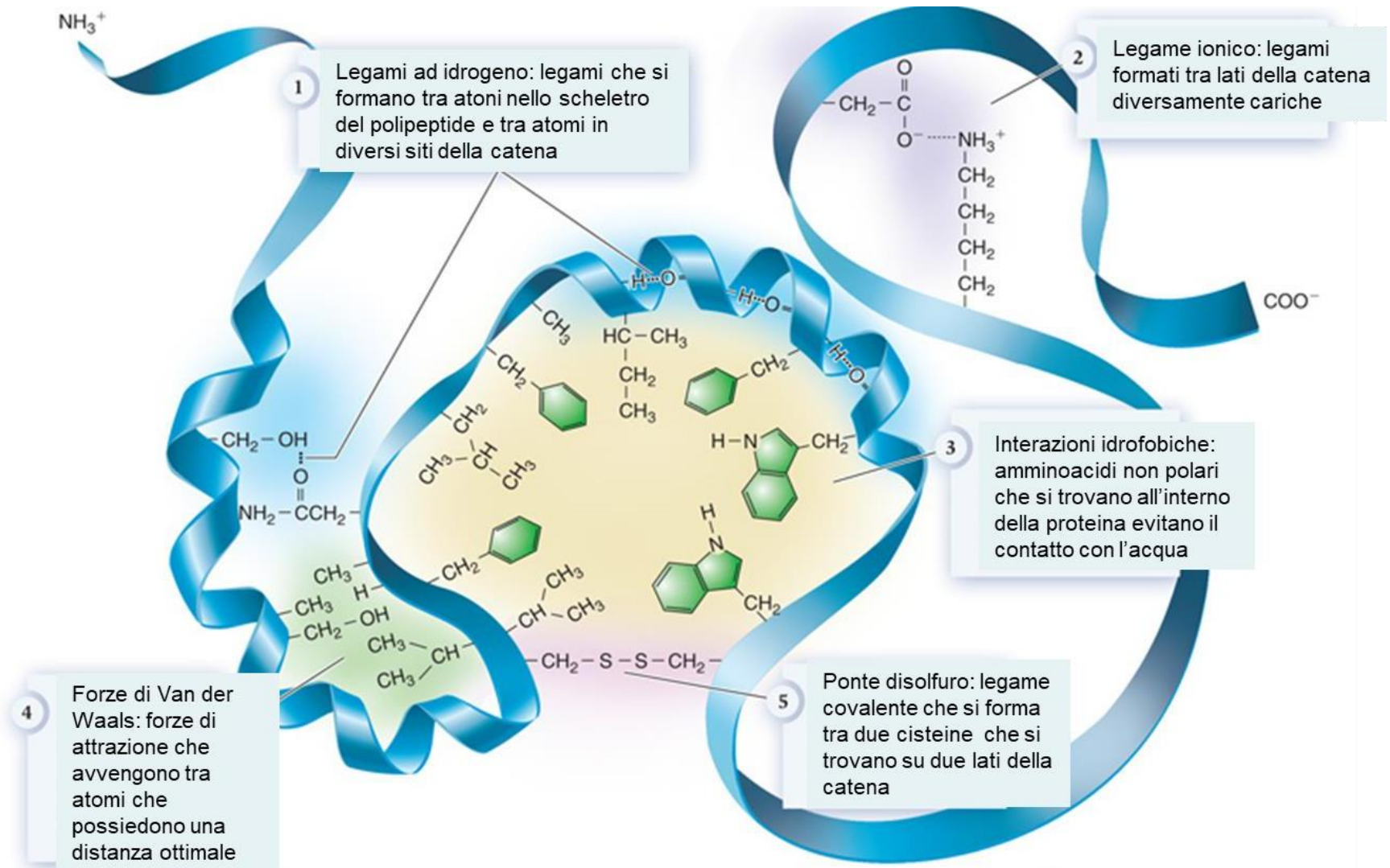


The Code of Life: Modulating gene expression



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Proteine – Interazioni tra amminoacidi



Proteine – Strutture dei peptide e polipeptidi

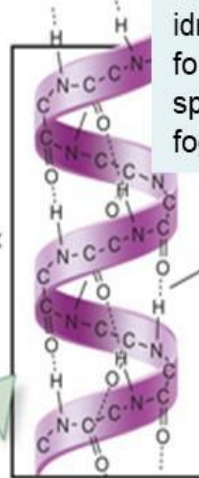
Struttura primaria: la sequenza lineare degli aminoacidi



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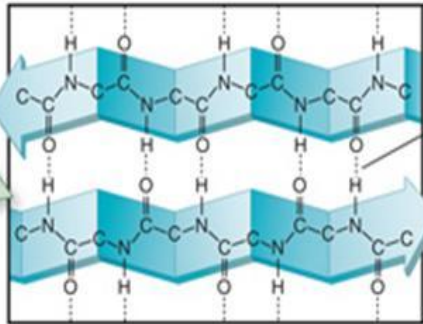
Struttura secondaria: alcune sequenze di aminoacidi formano legami ad idrogeno che provocano la formazione di strutture a spirale (α -elica) oppure a foglietto ripiegato (β -sheet)

α helix



H bond

β sheet



H bond

required for reproduction or display.

Struttura terziaria: strutture secondaria e regioni "random coiled" assumono una forma tri-dimensionale

Random coiled regions

Struttura quaternaria: due o più polipeptidi possono legarsi tra loro per formare una proteina funzionale



Proteine – Funzioni

Enzymatic proteins

Function: Selective acceleration of chemical reactions

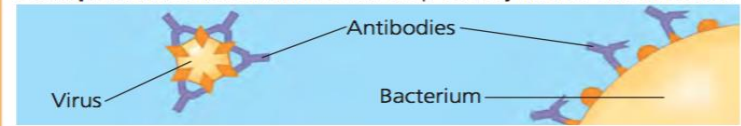
Example: Digestive enzymes catalyze the hydrolysis of bonds in food molecules.



Defensive proteins

Function: Protection against disease

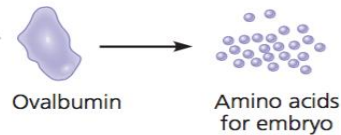
Example: Antibodies inactivate and help destroy viruses and bacteria.



Storage proteins

Function: Storage of amino acids

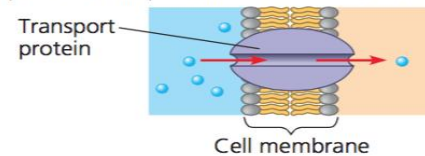
Examples: Casein, the protein of milk, is the major source of amino acids for baby mammals. Plants have storage proteins in their seeds. Ovalbumin is the protein of egg white, used as an amino acid source for the developing embryo.



Transport proteins

Function: Transport of substances

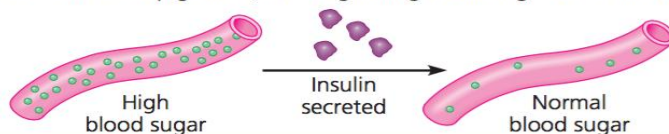
Examples: Hemoglobin, the iron-containing protein of vertebrate blood, transports oxygen from the lungs to other parts of the body. Other proteins transport molecules across cell membranes.



Hormonal proteins

Function: Coordination of an organism's activities

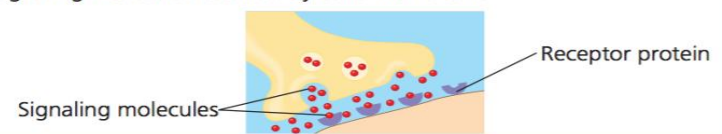
Example: Insulin, a hormone secreted by the pancreas, causes other tissues to take up glucose, thus regulating blood sugar concentration.



Receptor proteins

Function: Response of cell to chemical stimuli

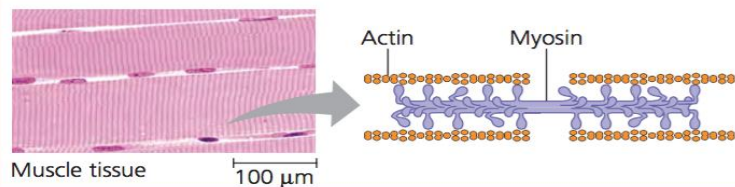
Example: Receptors built into the membrane of a nerve cell detect signaling molecules released by other nerve cells.



Contractile and motor proteins

Function: Movement

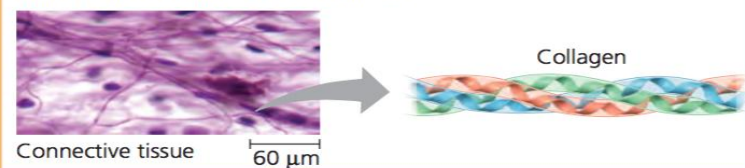
Examples: Motor proteins are responsible for the undulations of cilia and flagella. Actin and myosin proteins are responsible for the contraction of muscles.



Structural proteins

Function: Support

Examples: Keratin is the protein of hair, horns, feathers, and other skin appendages. Insects and spiders use silk fibers to make their cocoons and webs, respectively. Collagen and elastin proteins provide a fibrous framework in animal connective tissues.



The Code of Life: Modulating gene expression

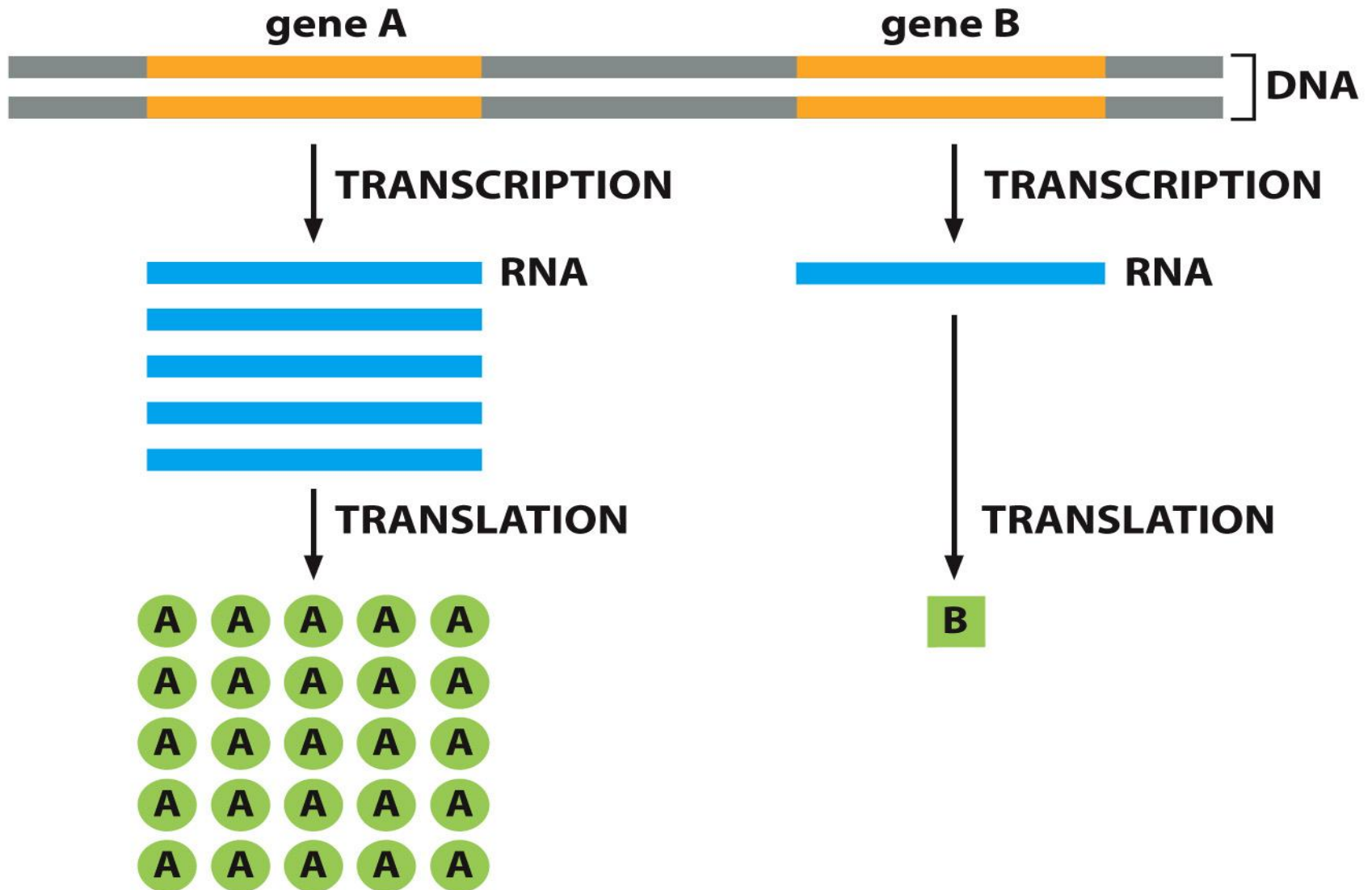


Figure 6-3 Molecular Biology of the Cell 6e (© Garland Science 2015)

The Code of Life: Modulating gene expression

- Protein synthesis may be induced by increasing:

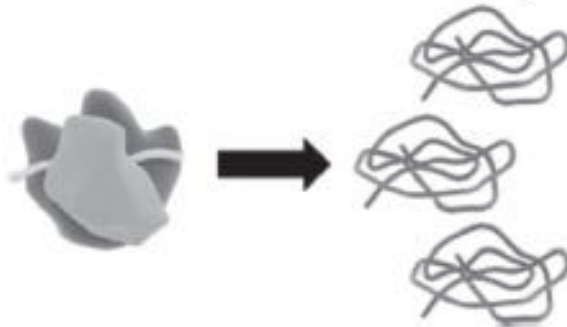
Translational Efficiency

↑ Rate of translation/ribosome = ↑ Proteins

Normal ribosomal translation



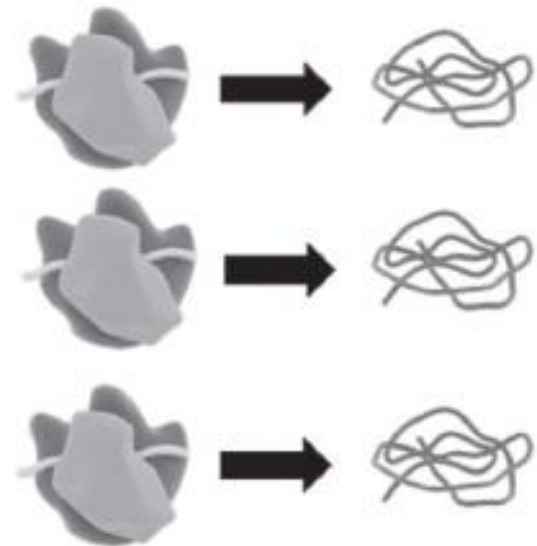
Increased translational efficiency



Translational capacity

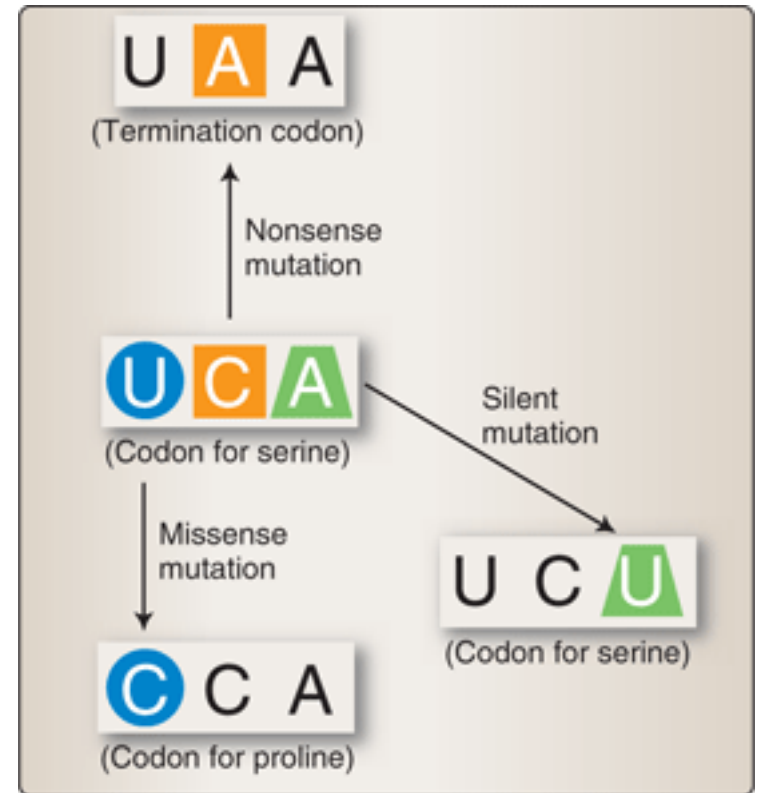
↑ Ribosomes = ↑ Proteins
→ ribosome biogenesis

Increased translational capacity



The Code of Life: Altering the nucleotide sequence

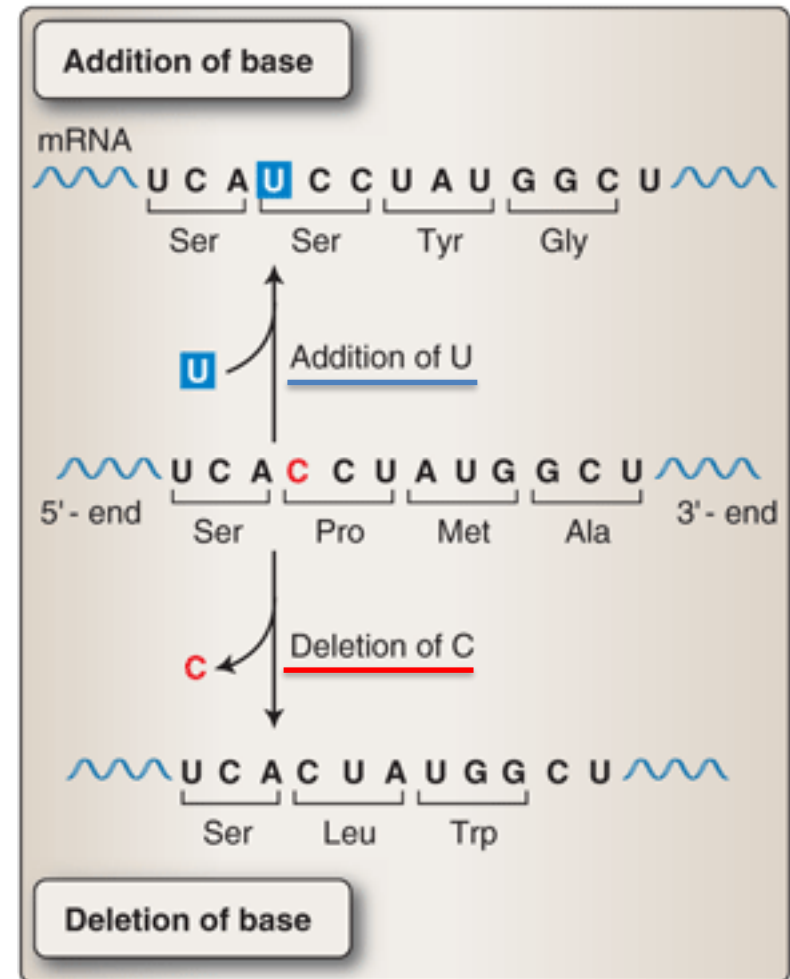
- Alterations in the codifying **nucleotide sequence** may alter the protein structure and function
- This may be at the base:
 - **Individual variants** (if present in more than >1% are called **polymorphisms**)
 - **Pathologies**
- **Silent mutation:** The codon containing the changed base may **code for the same amino acid**.
- **Missense mutation:** The codon containing the changed base may **code for a different amino acid**. For example, if the serine codon UCA is given a different first base "C" to become CCA, it will code for a different amino acid, in this case, proline.
- **Nonsense mutation:** The codon containing the changed base may **become a termination codon**. For example, if the serine codon UCA is given a different second base "A" to become UAA, the new codon causes termination of translation at that point and the production of a **truncated protein**.



The Code of Life: Altering the nucleotide sequence

Additional mutations:

- Frame-shift mutations:** If one or two nucleotides are either deleted from or added to the coding region the reading frame is altered. This can result in a product with a **radically different amino acid sequence** or a truncated product due to the **creation of a termination codon downstream** of the mutation. If three nucleotides are added, a new amino acid is added to the peptide, or if three nucleotides are deleted, an amino acid is lost. In these instances, the reading frame is not affected.
- Trinucleotide repeat expansion:** Occasionally, a sequence of three bases that is repeated in tandem will become amplified in number so that too many copies of the triplet occur.
- Splice site mutations:** Mutations at splice sites can alter the way in which introns are removed from the pre-mRNA molecules, producing aberrant proteins.
- Mutations can affect also **non-coding portion of the genome**



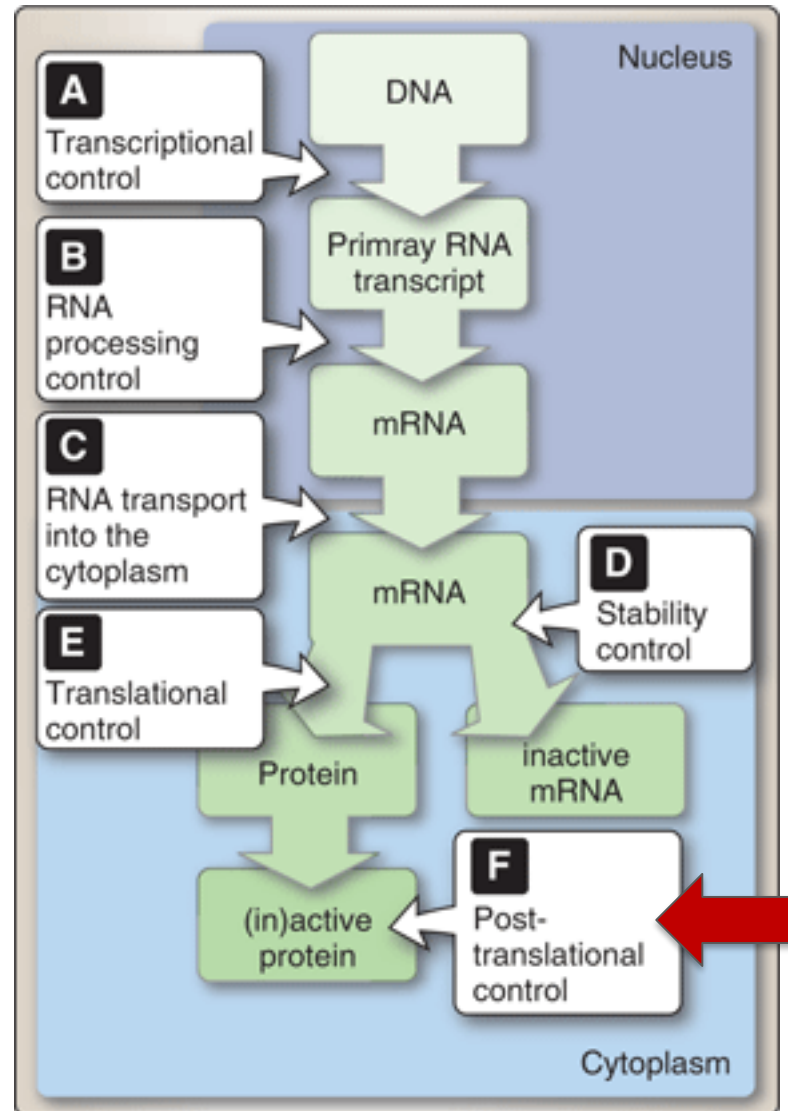
The Code of Life: Bacterial versus Eukaryotic Ribosomes

Table 22-1 Properties of Bacterial and Eukaryotic Cytoplasmic Ribosomes

Source	Size of Ribosomes		Subunit	Subunit Size		Subunit Proteins	Subunit RNA	
	S Value*	Mol. Wt.		S Value	Mol. Wt.		S Value	Nucleotides
Bacterial cells	70S	2.5×10^6	Large	50S	1.6×10^6	34	23S	2900
							5S	120
			Small	30S	0.9×10^6	21	16S	1540
Eukaryotic cells	80S	4.2×10^6	Large	60S	2.8×10^6	About 46	25–28S	≤4700
							5.8S	160
			Small	40S	1.43×10^6	About 32	5S	120
							18S	1900

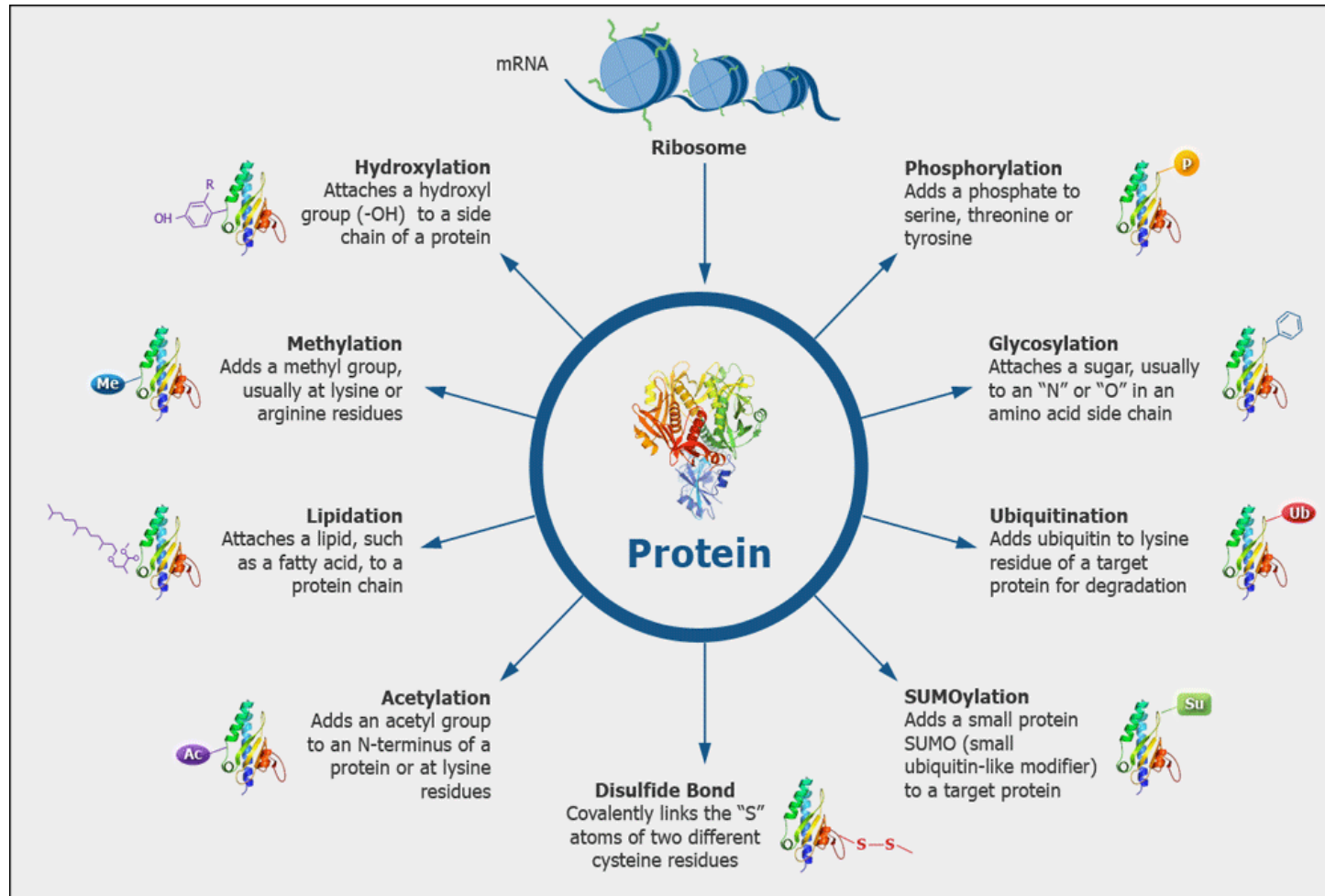
*If you are surprised that the S values of the subunits do not add up to that of the whole ribosome, recall that an S value is a measure of the velocity at which a particle sediments upon centrifugation and is only indirectly related to the mass of the particle.

The Code of Life: Modulation of Protein Function



The Code of Life: Post-Translational Modifications

Le modificazioni post-traduzionali delle protein sono generalmente mediate da specifiche proteine mentre altre sono specializzate nella loro successiva rimozione



The Code of Life: Post-Translational Modifications

TABLE 3–3 Some Molecules Covalently Attached to Proteins Regulate Protein Function

Modifying group	Some prominent functions
Phosphate on Ser, Thr, or Tyr	Drives the assembly of a protein into larger complexes (see Figure 15–11)
Methyl on Lys	Helps to create distinct regions in chromatin through forming either <u>mono-, di-, or trimethyl lysine in histones</u> (see Figure 4–36)
Acetyl on Lys	Helps to activate genes in chromatin by <u>modifying histones</u> (see Figure 4–33)
Palmityl group on Cys	This fatty acid addition <u>drives protein association with membranes</u> (see Figure 10–18)
N-acetylglucosamine on Ser or Thr	Controls <u>enzyme activity</u> and gene expression in glucose homeostasis
Ubiquitin on Lys	<u>Monoubiquitin</u> addition regulates the transport of membrane proteins in vesicles (see Figure 13–50)
	A <u>polyubiquitin chain</u> targets a <u>protein for degradation</u> (see Figure 3–70)
Ubiquitin is a 76-amino-acid polypeptide; there are at least 10 other ubiquitin-related proteins in mammalian cells.	

Table 3-3 Molecular Biology of the Cell 6e (© Garland Science 2015)

The Code of Life: Post-Translational Modifications

Epigenetic control of gene expression through chromatin modification

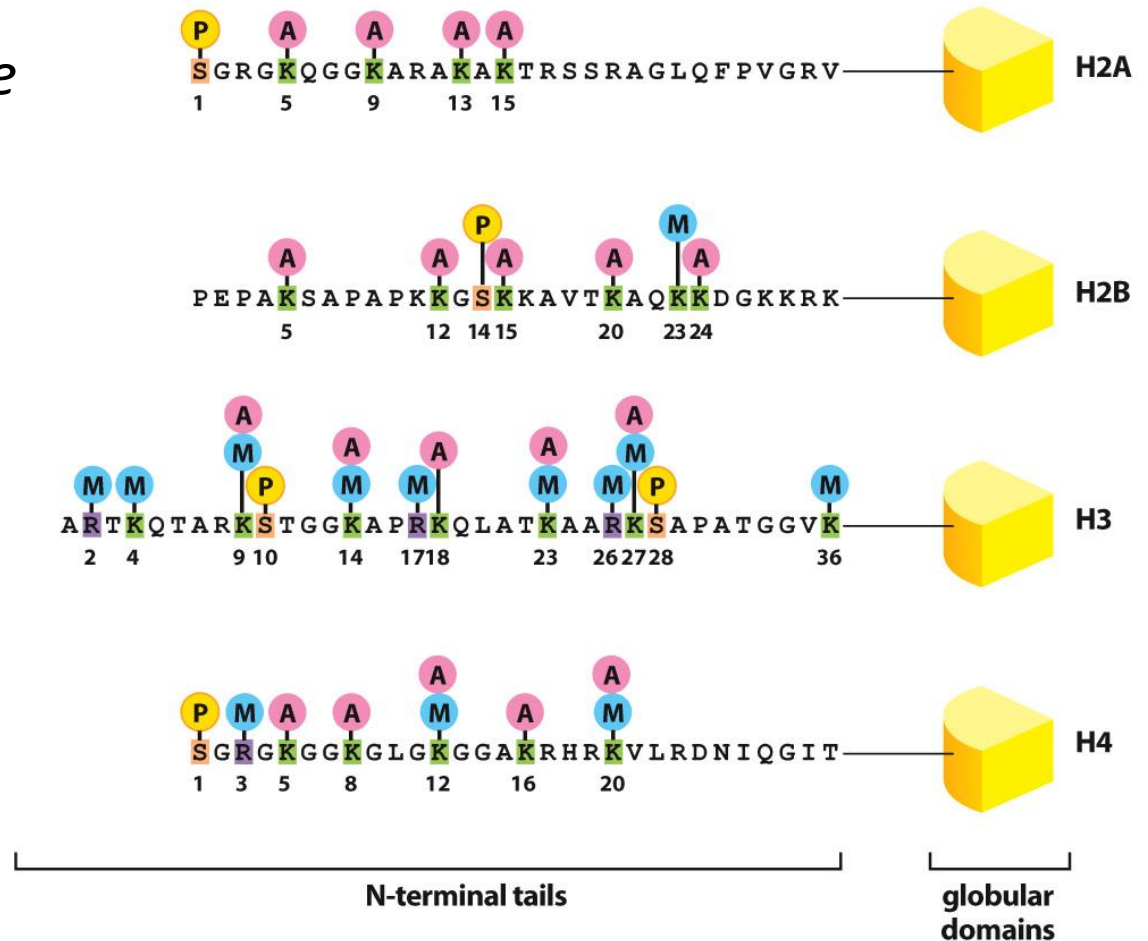
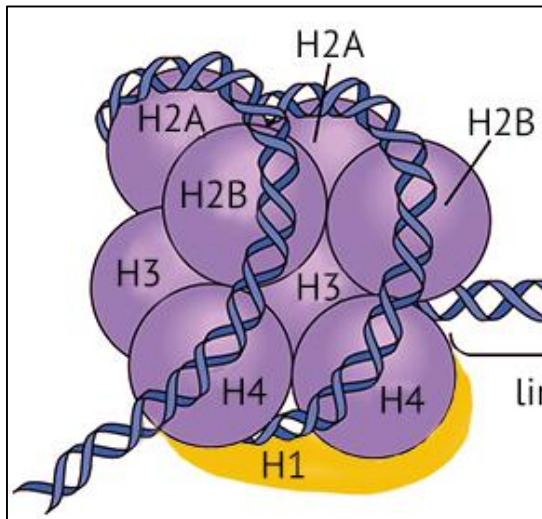
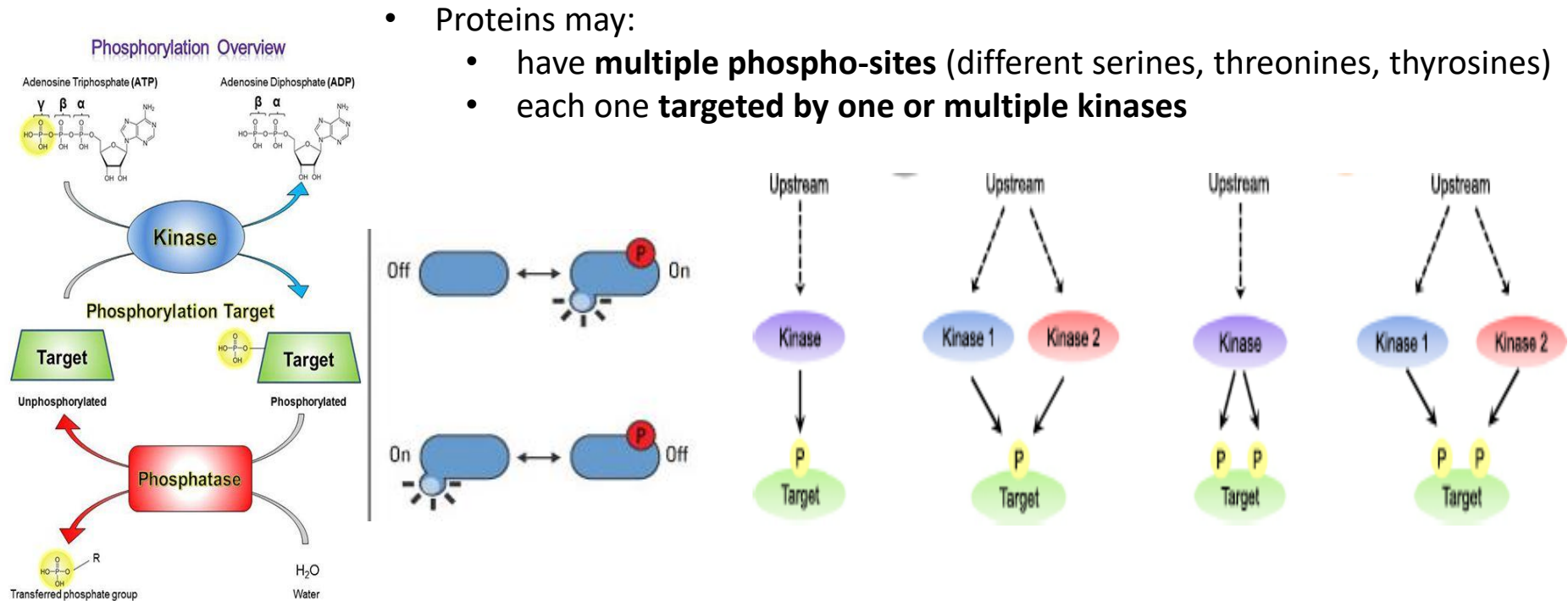


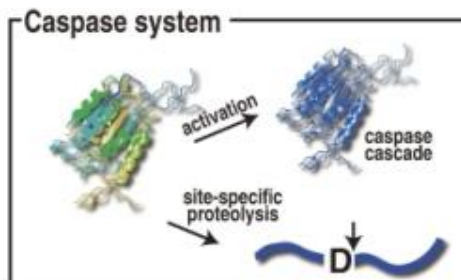
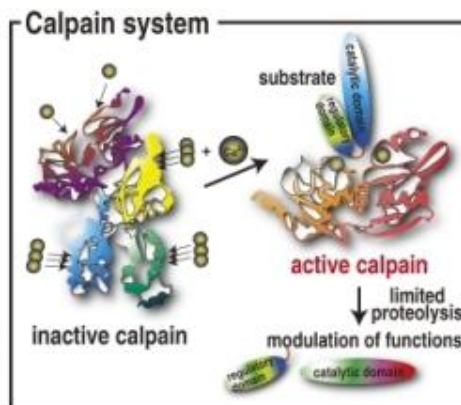
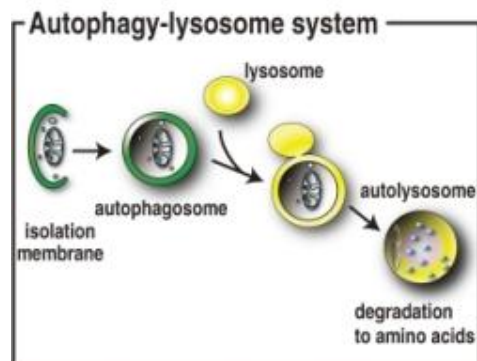
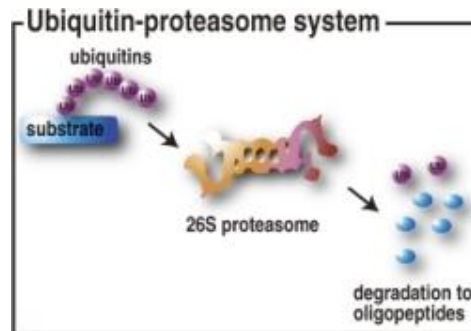
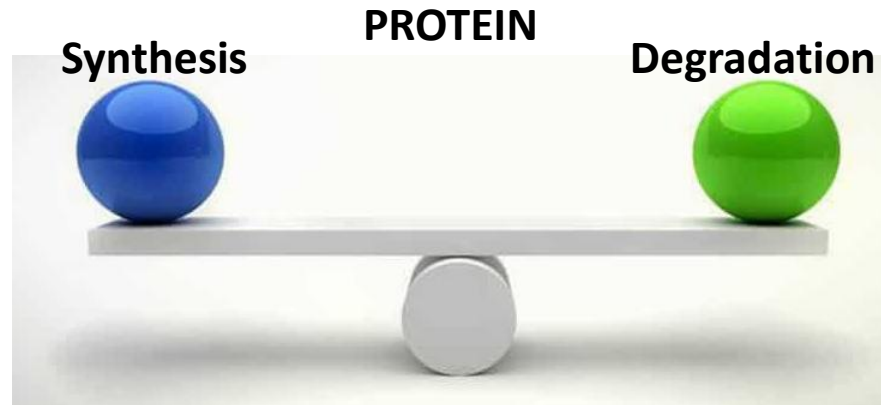
Figure 4-34b Molecular Biology of the Cell 6e (© Garland Science 2015)

The Code of Life: Post-Translational Modifications

- **Phosphorylation** is a reversible post-translational modification → Kinase vs. Phosphatase
- The **human genome encodes 518 protein kinases**:
 - **428** are known or predicted to **phosphorylate serine and threonine** residues
 - **90** are members of the **tyrosine kinase** family.
- By contrast, there are only ~ **200 phosphatases**, targeting phosphorylated proteins or lipids.
- Kinases and Phosphatases may target multiple phospho-sites on multiple proteins.



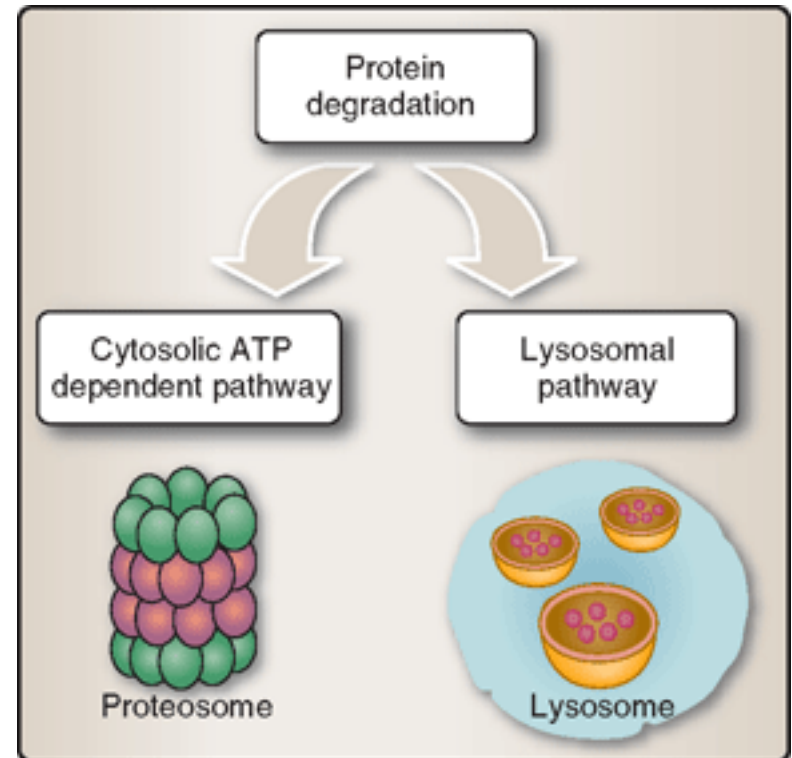
The Code of Life: Post-Translational Modifications



The Code of Life: Post-Translational Modifications

Degradazione Proteica

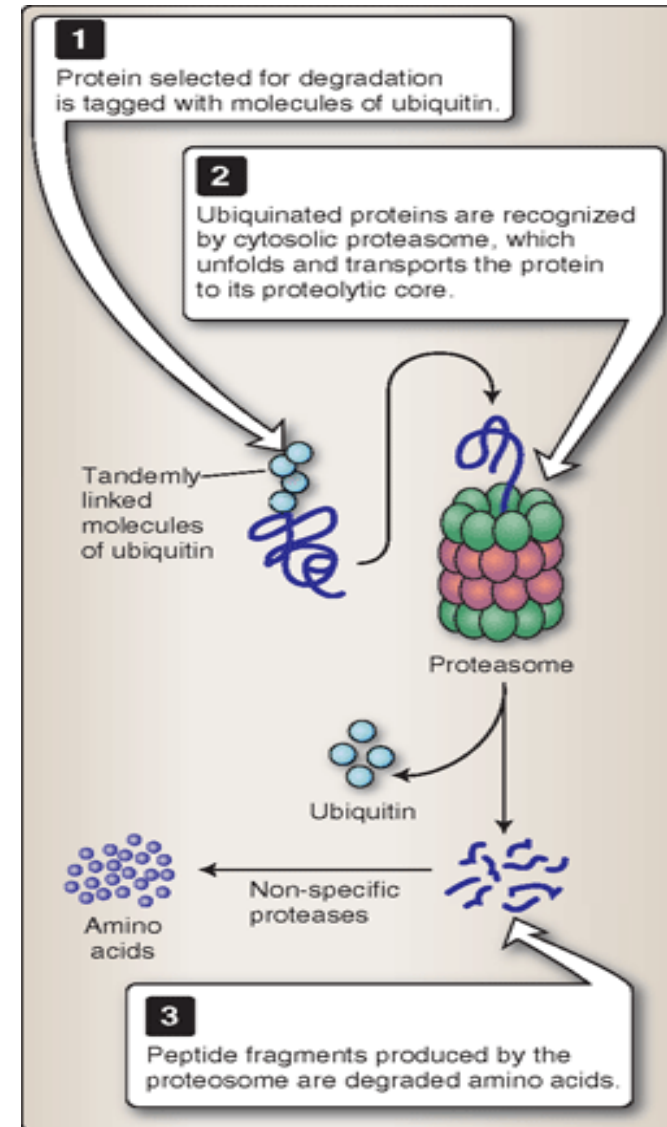
- **Protein's degradation** play an important role in **functional modulation**. Moreover permits the cell a constant **supply of amino acids** as well as **prevents the accumulation of abnormal proteins**.
- Two major degradation pathways:
 - **Proteosome pathway: ATP-dependent protein degradation pathway** which operates in the cytosol. Particularly **active for short-lived proteins** with half-life in seconds and minutes. This system is also important for the **degradation of defective proteins**, proteins that have suffered damage, and key **regulatory enzymes** of metabolic pathways.
 - **Lysosomal pathway**. Lysosomes form important **organelles** as they have the capacity to degrade a variety of biological molecules. Generally degrade **proteins with long half-lives**, such as **structural** or membrane proteins, **extracellular proteins**



The Code of Life: Post-Translational Modifications

Proteasome: The 26S proteasome is a large complex (made up of **about 60 protein subunits**) and resembles a **large cylinder**. It contains a **central core (20S)** and a **19S regulatory particle** at either end:

- 19S particle contains **several ATPases** and other enzymes and is important for **recognition** and binding of polyubiquitinated proteins, **removal of ubiquitin**, **unfolding the protein** substrate, and **translocation into the central core**.
- Proteins are then **hydrolyzed within the central core** into **smaller peptides**, which are **further degraded by cytosolic peptidases**.
- The ATP-dependent pathway involves the protein **ubiquitin**, a highly conserved **protein containing 76 amino acids** which, as the name suggests, is ubiquitous in the eukaryotic kingdom. Proteins destined for destruction are **tagged by multiple covalent attachment** of ubiquitin (**→ polyubiquitination**).



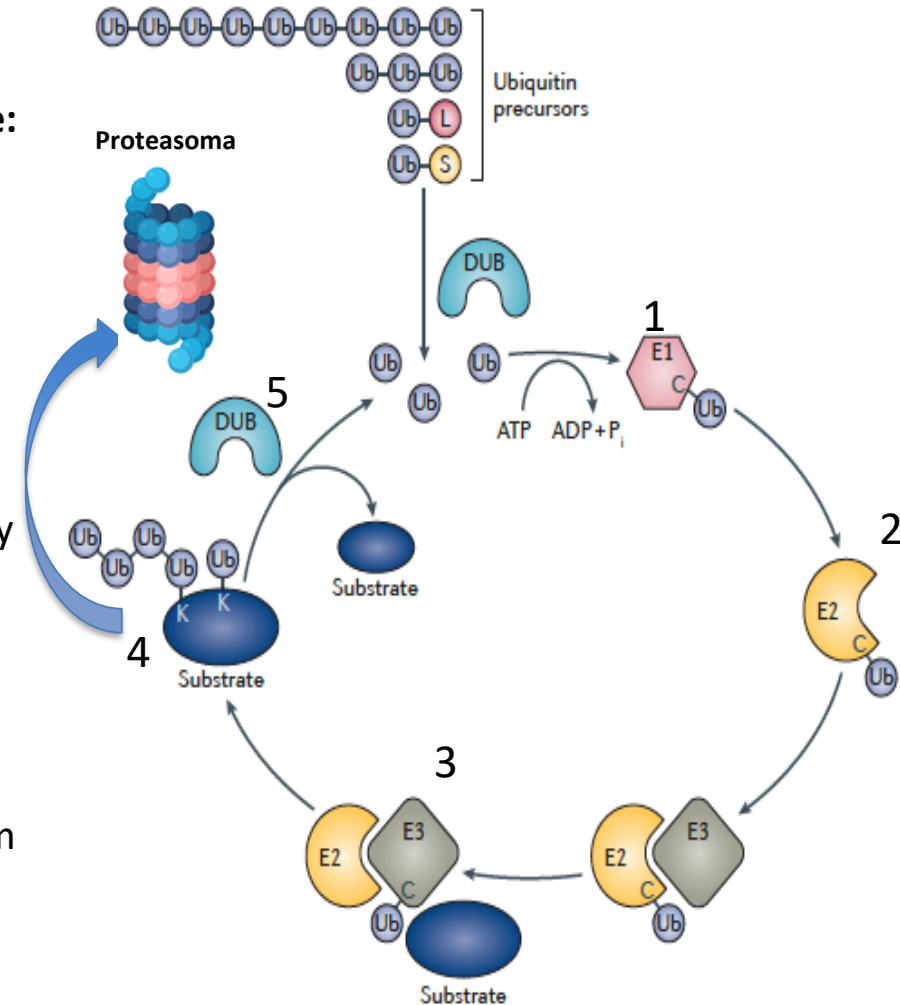
The Code of Life: Post-Translational Modifications

Different classes of enzymes (**Ub-ligases**) assists or directly **catalyse the transfer of ubiquitin to a substrate**:

1. E1 enzyme activates ubiquitin in an ATP-dependent manner, resulting in a covalent thioester linkage between ubiquitin and the E1 cysteine residue.
2. Ubiquitin is then transferred to an E2 enzyme, forming a thioester linkage with the catalytic cysteine.
3. An E3 ligase, catalyze substrate ubiquitination usually via a lysine side chain (K).
4. In subsequent rounds, ubiquitin can be conjugated to the N-terminal amino group or lysines on ubiquitin itself to form chains.

Deubiquitylating enzymes (DUBs) remove ubiquitin from substrates or ubiquitin precursors to generate free ubiquitin

~50 E2 enzymes, ~600 E3 ligases, ~ 90 DUBs



Harrigan et al 2018 Nature Review Drug Discovery

Mutations in genes encoding ubiquitin ligases/deubiquitinases are associated to human diseases

Table 1 | DUBs connected with human diseases

Process Targeted	DUB	Target	Rationale	Disease association	Refs
Oncology					
Proteasome	PSMD14	Many	General protein turnover	Liver cancer	41
	USP14			Lung and ovarian cancers	47,48,240
	UCHL5			Oesophageal and ovarian cancers	56,240,241
DNA repair	USP1	FANCD2, PCNA	* Fanconi anaemia pathway * Translesion synthesis	Osteosarcoma	73
	USP4	CTIP	Homologous recombination	Lung, breast and liver cancers	242–244
	USP11	PALB2		Breast cancers	245
	USP9X	Claspin	Replication checkpoint	Sarcoma and tumours of the colon, cervix, kidney, breast, prostate and brain	21
Oncogenes and tumour suppressors	ATXN3	p53, HDM2	Promotes p53-mediated apoptosis	p53-expressing tumours	102
	CYLD	NF-κB	Unclear	* Mutated in cylindromatosis and multiple myeloma * Reduced expression in colon and liver cancers and melanoma	23, 280–283
	UCHL1	AKT		Osteosarcoma, myeloma and tumours of the colon, breast, lung and kidney	252–258
	USP6	–		Translocated in aneurysmal bone cysts	284
	USP7	p53, HDM2	HDM2-overexpressing tumours	Leukaemia and ovarian and lung cancers	246–249
	USP8	EGFR	Regulates recycling of receptor tyrosine kinases, including EGFR	* Lung cancer * Mutated in Cushing syndrome	18,19, 107,285
	USP15	Type I TGFβ receptor, R-SMADs	Regulation of TGFβ signalling	Glioblastoma and breast and ovarian cancers	22,112
	USP20	HIF1α	Sensitizes hypoxic tumour cells	–	290
	USP28	FBW7, MYC, JUN, Notch	APC-driven cancers	Colorectal and ovarian cancers	250,251
	BAP1	Histone H2A, HCF1	Epigenetic deregulation of tumours	Uveal melanoma, sporadic melanoma, mesothelioma and kidney cancer	286–289
	USP22	Histone H2A		Colorectal, breast, oesophageal, lung and pancreatic cancers	127–130, 259,260
CNS disorders					
Neurodegeneration	ATXN3	Parkin	Counteracts Parkin autoubiquitylation	Expansion of CAG trinucleotide repeats causes Machado–Joseph disease	151
	USP7	α-Synuclein, REST	* Antagonizes ubiquitylation of α-synuclein * Regulates REST signalling and neuronal differentiation	–	149
	USP8	Parkin, K6-linked Ubiquitin chains	* Regulates mitophagy by removing ubiquitin from Parkin * Regulates TRKA levels in an NGF-dependent manner	–	148, 263–265
	USP14	Proteasome substrates	Increased clearance of proteins involved in neurodegeneration (Iau or ATXN3)	Mutations cause ataxia	44,262
	USP15	–	Opposes Parkin-mediated mitophagy	Glioblastoma	143,150
	USP30	Ubiquitin conjugates at mitochondrial surface, Parkin	Mitochondrial dysfunction, mitophagy	–	142,144, 146,261
Down syndrome	USP16	Histone H2A	Antagonizes self-renewal and/or senescence in Down syndrome	–	158–160, 266

Table 1 (cont.) | DUBs connected with human diseases

Process Targeted	DUB	Target	Rationale	Disease association	Refs
Inflammation, immunity and infectious disease					
Negative regulation of the immune response	A20	NEMO, RIPK1, TRAF6	Inhibits NF-κB signalling	Expression levels regulated by TNFα, IL-1β and LPS	167, 273,274
	CYLD	RIG1, TBK1, IKKε		–	170,171
	OTULIN	RIPK1, RIPK2, NEMO		–	172,173
	USP18	–	* Functions in haematopoietic cell differentiation * Removes ISG15 conjugates * Negative feedback regulator of type I IFN signalling	Expression regulated by IFNγ	268, 269,291
	USP25	RIG1, TRAF2, TRAF3, TRAF6	* Negatively regulates IL-17-triggered signalling * Negatively regulates virus-induced type I IFN production * Positive feedback regulation of innate immune responses against RNA and DNA viruses	Expression regulated by IFN and IRF7	270–272, 292,303
T _H responses	USP7	FOXP3	* Stabilizes FOXP3 in T _H cells * Negative regulator of TNFα-stimulated NF-κB activity	Expressed and regulated upon viral infections in B and T cells	34,267,293
	USP21	FOXP3	Stabilizes FOXP3 in T _H cells	–	131
T _H 1 and T _H 17 responses	CEZANNE	ZAP70	* Positive regulator of T cell receptor signalling * Binds to and deubiquitylates ZAP70	–	193
	TRABID	JMJD2D	Positive regulator of IL-22 and IL-23 cytokine production	–	191
	USP4	RORγt, RIG1, TAK1	* Stabilizes RORγt in T _H 17 cells * Positively regulates RIG1-mediated antiviral response	Highly expressed in CD4 ⁺ T cells from patients with rheumatic heart disease	189, 275,276, 294
	USP10	T-bet	* Negative regulator of TLR-IL-1R signalling * Targets TAK1 to downregulate TNFα-induced NF-κB activation	Highly expressed in PBMCs from patients with asthma	196
	USP17	RORγt, RIG1, IL-33	* Positive regulator of RORγt in T _H 17 cells * Regulates virus-induced type I IFN signalling * Regulates the stability and nuclear function of IL-33	–	277–279, 295
	USP18	TAK1–TAB1 complex	Regulates TAK1–TAB1 interaction required for T _H 17 differentiation	Expression induced by cytokines	192

APC, adenomatous polyposis coli protein; ATXN3, ataxin 3; BAP1, BRCA1-associated protein 1; CEZANNE, cellular zinc finger anti-NF-κB protein; CNS, central nervous system; CYLD, cylindromatosis; CTIP, C-terminal-binding protein-interacting protein; EGFR, epidermal growth factor receptor; FANCD2, Fanconi anaemia group D2 protein; FBW7, F-box and WD40 domain-containing protein 7; FOSL1, forkhead box protein P3; HCF1, host cell factor 1; HDM2, human double minute 2; HIF1α, hypoxia-inducible factor 1α; IKKε, IκB kinase-ε; IFN, interferon; IL, interleukin; IRF7, interferon regulatory factor 7; ISG15, IFN-stimulated gene 15; JMJD2D, Jumoni domain-containing protein 2D; LPS, lipopolysaccharide; NEMO, NF-κB essential modulator; NF-κB, nuclear factor-κB; NGF, β-nerve growth factor; PALB2, partner and localizer of BRCA2; PBMC, peripheral blood mononuclear cell; PCNA, proliferating cell nuclear antigen; REST, RE1-silencing transcription factor; RIG1, retinoic acid-inducible gene 1-like receptor 1; RIPK, receptor-interacting serine/threonine-protein kinase; RORγt, retinoid-related orphan receptor γt; R-SMAD, receptor-regulated SMAD; TAB, TAK1-binding protein; TAK1, TGFβ-activated kinase 1; TBK1, TANK-binding kinase 1; TGFβ, transforming growth factor-β; T_H cells, T helper cells; TLR, Toll-like receptor; TNFα, tumour necrosis factor-α; TRAF6, TNF receptor-associated factor 6; T_H cells, regulatory T cells; TRKA, tropomyosin-related kinase A; UCHL1, ubiquitin carboxy-terminal hydrolase-like; USP, ubiquitin-specific protease; ZAP70, 70kDa ζ-chain-associated protein.

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